

SURFACE CHEMISTRY

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THEORY

1. GENERAL INTRODUCTION

Surface Chemistry is that branch of chemistry which deals with the study of the phenomena occurring at the surface or interface, i.e., at the boundary separating two bulk phases.

The two bulk phases can be pure compounds or solutions. The interface is represented by putting a hyphen or a slash between the two bulk phases involved, e.g., solid-liquid or solid/liquid. No interface exists between gases as they are completely miscible. Important phenomena occur at the interface, e.g., dissolution, crystallisation, corrosion, heterogeneous catalysis, electrode processes, etc.

2. ADSORPTION

2.1 Definition

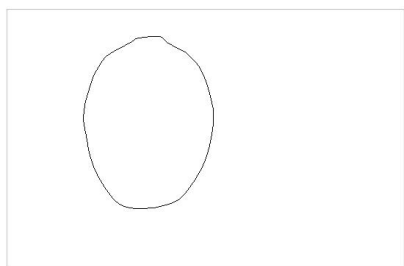
The phenomenon of attracting and retaining the molecules of a substance on the surface of a liquid or solid resulting in to higher concentration of the molecules on the surface is called **adsorption**. Adsorption of gases at metal surface is called occlusion

2.2 Adsorbate and Adsorbent

The substance which gets adsorbed on any surface is called **adsorbate** for example, if a gas gets adsorbed on to the surface of a solid, then the gas is termed as the adsorbate.

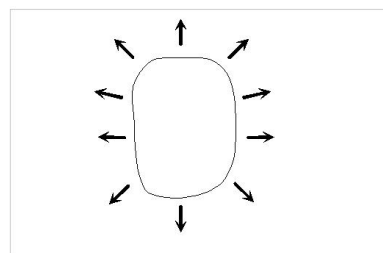
The substance on the surface of which adsorption takes place is called **adsorbent**.

Adsorbent may be a solid or a liquid. Metal powders, powdered charcoal, animal charcoal silica powder etc. are commonly used as adsorbents.



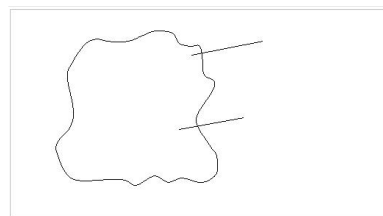
2.3 Desorption

The removal of the adsorbed substance from a surface is called **desorption**. This can be done by heating or reducing the pressure of the system.



2.4 Absorption

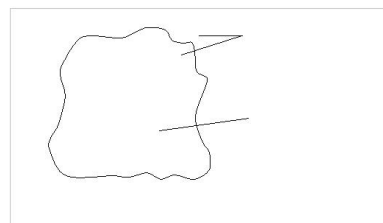
When the molecules of a substance are uniformly distributed throughout the body of a solid or liquid, this phenomenon is called **absorption**.



2.5 Sorption

The phenomenon in which adsorption and absorption occur simultaneously is called **sorption**. Dyes are absorbed as well as adsorbed in cotton fibre.

Adsorption is instantaneous *i.e.* a fast process while absorption is a slow process.



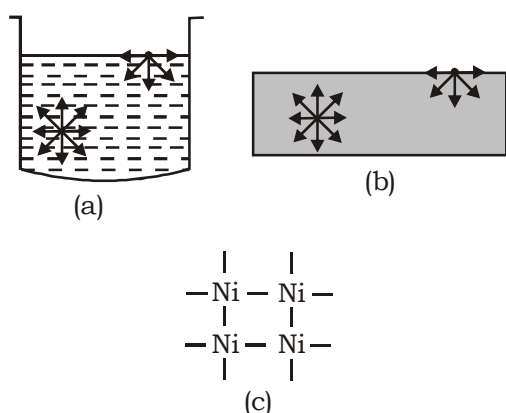
2.6 Difference between Adsorption and Absorption

Main points of difference between adsorption and absorption are given below.

DIFFERENCE BETWEEN ADSORPTION AND ABSORPTION

Adsorption	Absorption
1. It is a surface phenomenon. 2. In it, the substance is only retained on the surface and does not go into the bulk or interior of the solid or liquid. 3. In it the concentration of the adsorbed molecules is always greater at the free phase. 4. It is rapid in the beginning and slows down to attain equilibrium Examples (i) Water vapours are adsorbed by CaCl_2. (ii) NH_3 is adsorbed by charcoal. (v) Decolourisation of sugar solution by activated or animal charcoal. (vi) Ink is adsorbed by blotting paper.	1. It concerns with the whole mass of the absorbent. 2. It implies that a substance is uniformly distributed, through the body of the solid or liquid. 3. In it the concentration is low. 4. It occurs at the uniform rate. Examples (i) Water vapours are absorbed by anhydrous silica gel. (ii) NH_3 is absorbed in water forming NH_4OH.

2.7 Mechanism of adsorption



Molecules at the surface experiencing a net inward force of attraction in case of

(a) liquid (b) solid (c) metal with free valencies

Only the surface atoms of an adsorbent play an active role in adsorption. These atoms possess unbalanced forces of various types such as, Vander Waal's forces and chemical bond forces (free valencies).

Thus, due to residual or unbalanced inward forces of attraction or free valencies at the surface, liquids and solids have the property to attract and retain the molecules of a gas or a dissolved substance on to their surface.

2.8 Characteristics of adsorption

2.8.1 Adsorption refers to the existence of a higher concentration of any particular component at the surface of a liquid or a solid phase.

2.8.2 Adsorption is invariably accompanied by evolution of heat, i.e. it is an exothermic process. In other words, **ΔH of adsorption is always negative.**

2.8.3 When a gas is adsorbed, the freedom of movement of its molecules becomes restricted. On account of it, decrease in the entropy of the gas after adsorption, i.e. **ΔS is negative.**

Adsorption is thus accompanied by decrease in enthalpy as well as decrease in entropy of the system and hence ΔG also decreases.

2.8.4 Adsorption is accompanied by decrease in the ΔG (free energy change) of the system when $\Delta G = 0$, **adsorption equilibrium is said to be established.**

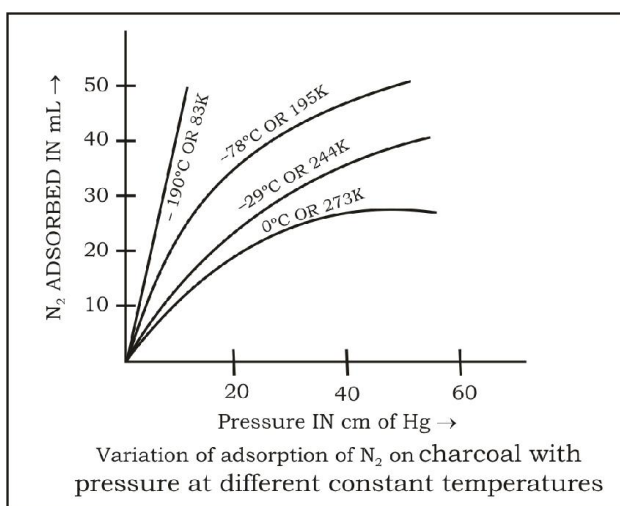
2.8.5 For a process to be spontaneous, the thermodynamic requirement is that **ΔG must be negative**, i.e. there is decrease in free energy. On the basis of **Gibb's Helmholtz equation**, $[\Delta G = \Delta H - T\Delta S]$, ΔG can be negative if ΔH has sufficiently high negative value and $T\Delta S$ has positive value.

Gas	H ₂	N ₂	CO	CH ₄	CO ₂	HCl	NH ₃	SO ₂
Volume adsorbed (mL)	4.7	8.0	9.3	16.2	48	72	181	380
Critical temp (K)	33	126	134	190	304	324	406	430
	Critical temperature increases →							
	Ease of liquefaction increases →							
	Adsorption increases →							

2.10.2 Surface area of the solid adsorbent : The extent of adsorption depends directly upon the surface area of the adsorbent, i.e. *larger the surface area of the adsorbent, greater is the extent of adsorption.*

2.10.3 Effect of pressure on the adsorbate gas :

- ♦ An increase in the pressure of the adsorbate gas increases the extent of adsorption.
- ♦ At low temperature, the extent of adsorption increases rapidly with pressure.
- ♦ At low pressure, the extent of adsorption is found to be directly proportional to the pressure.
- ♦ At high pressure (closer to the saturation vapour pressure of the gas), the adsorption tends to achieve a limiting value.



2.10.4 Effect of temperature : As adsorption is accompanied by evolution of heat, so according to the **Le-Chatelier's principle**, the magnitude of adsorption should decrease with rise in temperature.

The amount of heat when one mole of the gas is adsorbed on the adsorbent is called the **heat of adsorption**.

2.11 Adsorption Isotherms

A mathematical equation which describes the relationship between pressure (p) of the gaseous adsorbate and the extent of adsorption at any fixed temperature is called **adsorption isotherms**.

Thus, if x g of an adsorbate is adsorbed on m g of the adsorbent, then

$$\text{Extent of adsorption} = \frac{x}{m}$$

2.11.1 Freundlich adsorption isotherm : Freundlich adsorption isotherm is obeyed by the adsorptions where the adsorbate forms a **monomolecular** layer on the surface of the adsorbent.

$$\frac{x}{m} = kP^{\frac{1}{n}} \quad \text{or} \quad \log \frac{x}{m} = \log k + \frac{1}{n} \log P$$

where, x is the weight of the gas adsorbed by m gm of the adsorbent at a pressure P , thus x/m represents the amount of gas adsorbed by the adsorbent per gm (unit mass), k and n are constant at a particular temperature and for a particular adsorbent and adsorbate (gas), n is always greater than one, indicating that the amount of the gas adsorbed does not increase as rapidly as the pressure.

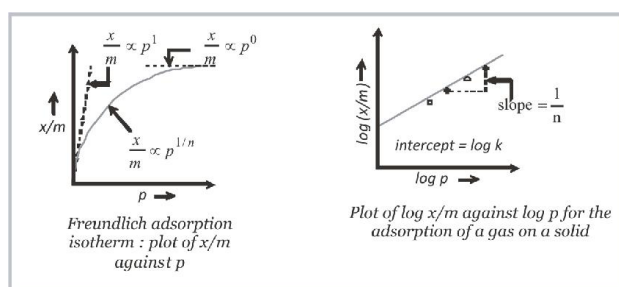
At low pressure, the extent of adsorption varies linearly

$$\text{with pressure } \frac{x}{m} \propto P^1$$

- ♦ At high pressure, it becomes independent of pressure

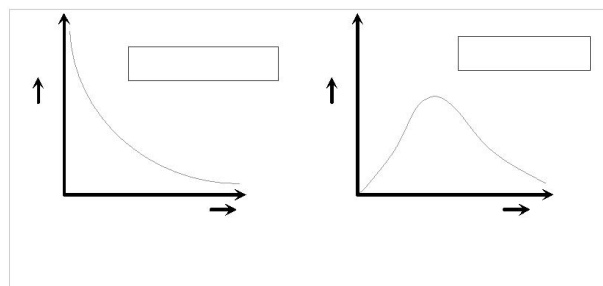
$$\frac{x}{m} \propto P^0$$

- ♦ At moderate pressure $\frac{x}{m}$ depends upon pressure raised to powers $\frac{x}{m} \propto P^{\frac{1}{n}}$



2.12 Adsorption Isobar

A graph drawn between the amount of the gas adsorbed per gram of the adsorbent (x/m) and temperature ' t ' at a constant equilibrium pressure of adsorbate gas is known as adsorption isobar.



The physical adsorption isobar shows a decrease in x/m throughout with rise in temperature, the chemisorption isobar shows an initial increase with temperature and then the expected decrease. The initial increase is because of the fact that the heat supplied acts as activation energy required in chemisorption (like chemical reactions).

2.13 Adsorption from Solutions

Solid surfaces can also adsorb solutes from solutions. A few examples are :

- ♦ When litmus solution is shaken with charcoal, it becomes colour less because the dye of the litmus solution is adsorbed by charcoal.
- ♦ When the colourless Mg(OH)_2 is precipitated in the presence of magneson reagent (a blue coloured dye). it acquires blue colour because the dye is adsorbed on the solid precipitate.

The extent of adsorption from solution depends upon the concentration of the solute in the solution, and can be expressed by the **Freundlich isotherm**. The Freundlich adsorption isotherm for the adsorption from solution is, $\frac{x}{m} = kc^n$ where, x is the mass of the solute adsorbed, m is the mass of the solid adsorbent, c is the equilibrium concentration of the solute in the solution, n is a constant having value greater than one. k is the proportionality constant, (The value of k depends upon the nature of solid, its particle size, temperature, and the nature of solute and solvent etc.) The plot of x/m against c is similar to that Freundlich adsorption isotherm. The above equations may be written in the following form, $\log \frac{x}{m} = \log k + \frac{1}{n} \log c$ where c , is the equilibrium concentration of the solute in the solution.

2.14 Factors affecting adsorption from solution

The adsorption from solutions by solid adsorbents is found to depend upon the following factors :

2.14.1 Nature of the adsorbate and the adsorbent.

2.14.2 The adsorption decreases with **temperature**.

2.14.3 Greater the **surface area of the adsorbent** greater is the adsorption.

2.14.4 Concentration of the solute in the solution.

2.15 Positive and Negative Adsorption

In case of adsorption by solids from the solutions, mostly the solute is adsorbed on the surface of the solid adsorbent so that concentration of solute on the surface of the adsorbent is greater than in the bulk. However in some cases, the solvent from the solution may be adsorbed by the adsorbent so that the concentration of the solution increases as compared to initial concentration.

When the concentration of the adsorbate is more on the surface of the adsorbent than in the bulk, it is called positive adsorption. If the concentration of the adsorbate increases in the bulk after adsorption, it is called negative adsorption.

Blood solution + Conc. KCl solution $\rightarrow$ Positive adsorption

Blood solution + dilute KCl solution $\rightarrow$ Negative adsorption

2.16 Applications of Adsorption

The phenomenon of adsorption finds a number of applications. Important applications are given as follows.

2.16.1 Production of high vacuum : A bulk of charcoal cooled in liquid air is *connected* to a vessel which has already been exhausted as far as possible by a vacuum pump. The remaining traces of air are adsorbed by the charcoal to produce a very high vacuum.

2.16.2 In Gas masks : It is a device which consists of activated charcoal or a mixture of adsorbents. This apparatus is used to adsorb poisonous gases (e.g. Cl_2 , CO, oxide of sulphur etc.) and thus purify the air for breathing in coal mines.

2.16.3 For desiccation or dehumidification : Certain substances have a strong tendency to absorb water such as silica and alumina (Al_2O_3). These substances can be used to reduce/remove water vapours or moisture present in the air. Silica gel is also used for dehumidification in electronic equipment.

2.16.4 Removal of colouring matter from solution : Animal charcoal removes colours of solutions by adsorbing coloured impurities. It is also used as decolouriser in the manufacture of cane sugar.

2.16.5 Heterogeneous catalysis : Mostly heterogeneous catalytic reactions proceed through the adsorption of gaseous reactants on solid catalyst. For example,

- ◆ Finely powdered nickel is used for the hydrogenation of oils.
- ◆ Finely divided vanadium pentaoxide (V_2O_5) is used in the contact process for the manufacture of sulphuric acid.
- ◆ Pt , Pd are used in many industrial processes as catalyst.
- ◆ Manufacture of ammonia using iron as a catalyst.

2.16.6 Separation of inert gases : Due to the difference in degree of adsorption of gases by charcoal, a mixture of inert gases can be separated by adsorption on coconut charcoal at different low temperatures.

2.16.7 Softening of hard water :

- ◆ The hard water is made to pass through a column packed with zeolite (sodium aluminium silicate)

2.16.8 De-ionisation of water :

- ◆ For softening water can be de-ionised by removing all dissolved salts with the help of cation and anion-exchanger resin.

2.16.9 In curing diseases : A number of drugs are adsorbed on the germs and kill them or these are adsorbed on the tissues and heat them.

2.16.10 Cleaning agents : Soap and detergents get adsorbed on the interface and thus reduce the surface tension between dirt and cloth, subsequently the dirt is removed from the cloth.

2.16.11 Froth floatation process :

- ◆ A low grade sulphide ore is concentrated by separating it from silica and other earthy matter by this method.
- ◆ The finely divided ore is added to water containing pine oil and foaming agent.
- ◆ The air is bubbled through the mixture.
- ◆ The foam formed rises to the surface on which mineral particles wetted with oil are adsorbed while earthy matter settle down at the bottom.

2.16.12 In adsorption indicators : Surface of certain precipitates such as silver halide, have the property of adsorbing some dyes like eosin, fluorescein etc. In this case of precipitation titrations (for example $AgNO_3$ Versus $NaCl$) the indicator is adsorbed at the end point producing a characteristic colour on the precipitate.

2.16.13 Chromatographic analysis : The phenomenon of adsorption has given an excellent technique of analysis known as chromatographic analysis. The technique finds a number of applications in analytical and industrial fields. Chromatographic technique is based on differential adsorption of different constituents of a mixture.

2.16.14 In dyeing : Many dyes get adsorbed on the cloth either directly or by the use of mordants.

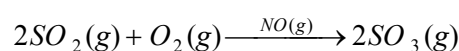
“Catalyst is a substance which speeds up and speeds down a chemical reaction without itself being used up at the end of the reaction and the phenomenon is known as catalysis.

3.1 Types of catalysis

Catalytic reactions can be broadly divided into the following types,

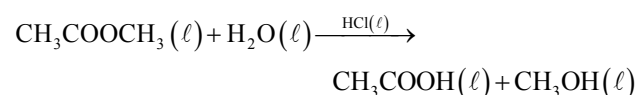
3.1.1 Homogeneous catalysis : When the reactants and the catalyst are in the same phase (*i.e.* solid, liquid or gas). The catalysis is said to be **homogeneous**. The following are some of the examples of homogeneous catalysis.

- ◆ Oxidation of sulphur dioxide into sulphur trioxide with oxygen in the presence of oxides of nitrogen as the catalyst in the lead chamber process.

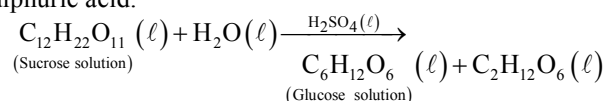


The reactants, products and catalyst all are in gaseous state *i.e.* same phase.

- ◆ Hydrolysis of methyl acetate is catalysed by H^+ ions furnished by hydrochloric acid .



- ◆ Hydrolysis of sugar is catalysed by H^+ ions furnished by sulphuric acid.



3.1.2 Heterogeneous catalysis : The catalytic process in which the reactants and the catalyst are in different phases is known as **heterogeneous catalysis**. Some of the examples of heterogeneous catalysis are given below.

- ◆ Oxidation of sulphur dioxide into sulphur trioxide in the presence of platinum metal or vanadium pentaoxide as catalyst in the contact process for the manufacture of sulphuric acid. The reactants are in gaseous state while the catalyst is in solid state. $2SO_2(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$

fails to catalyse the other reaction, different catalysts for the same reactant may give different products.

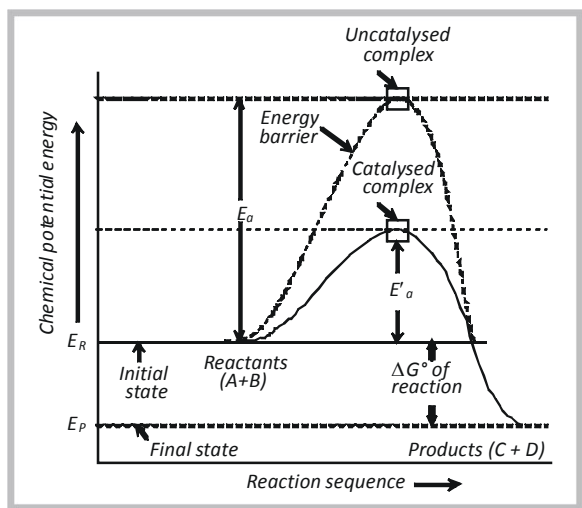
3.2.4 The catalyst can not change the position of equilibrium: The catalyst catalyses both forward and backward reactions to the same extent in a reversible reaction and thus has no effect on the equilibrium constant.

3.2.5 Catalytic promoters : Substances which themselves are not catalysts, but when mixed in small quantities with the catalysts increase their efficiency are called as **promoters** or **activators**.

3.2.6 Catalytic poisons : Substances which destroy the activity of the catalyst by their presence are known as **catalytic poisons**.

3.2.7 Change of temperature alters the rate of catalytic reaction as it does for the same reaction in absence of catalyst : By increasing the temperature, there is an increase in the catalytic power of a catalyst but after a certain temperature its power begins to decrease. A catalyst has thus, a particular temperature at which its catalytic activity is maximum. This temperature is termed as **optimum temperature**.

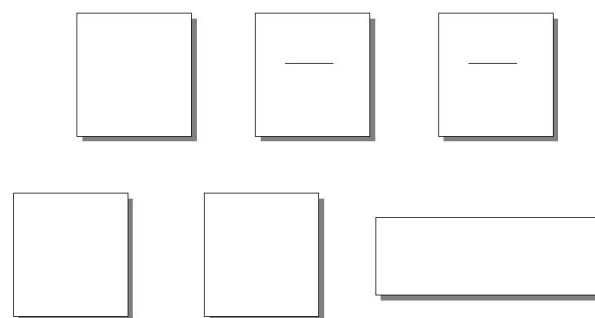
A positive catalyst lowers the activation energy



3.3 Theories of Catalysis

3.3.1 Collision Theory of Homogeneous Catalysis :

According to the collision theory, a reaction occurs on account of effective collisions between the reacting molecules. For effective collision, it is necessary that the molecules must possess a minimum amount of energy known as activation energy (E_a). After the collision molecules form an activated complex which dissociates to yield the product molecules. The catalyst provides a new pathway involving lower amount of activation energy. Thus,



3.3.2 Adsorption theory of Heterogeneous Catalysis :

Heterogeneous catalytic reactions generally proceed via adsorption of reactants on the surface of the catalyst.

Mechanism of such surface reactions may be explained in terms of **diffusion theory of catalysis**. This theory postulates the following sequence for gaseous reactions on a solid surface.

Step: (i) Diffusion of the reactants to the surface.

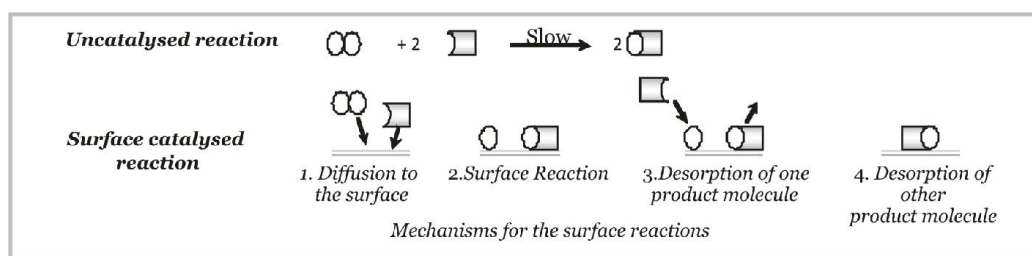
Step: (ii) Adsorption of the reactant molecules onto the surface.

Step: (iii) Actual chemical reaction on the surface.

Step: (iv) Desorption of the products from the surface.

Step: (v) Diffusion of the products away from the surface.

In generally, Step (iii) determines the rate of reaction. However step (ii) and (iv) may be rate determining.



3.3.3 Advantages of modern adsorption theory : This theory can explain the following :

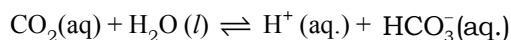
- Small quantity of the catalyst is sufficient because the catalyst is regenerated again and again.
- The catalyst takes part in the reaction but is produced back unchanged in mass and chemical composition at the end of the reaction.
- Catalytic poisons, if present, are preferentially adsorbed on the catalyst surface, thereby hindering the adsorption of the reactant molecules. Hence, they slow down the activity of the catalyst.

This theory, however, cannot explain satisfactorily the action of catalytic promoters.

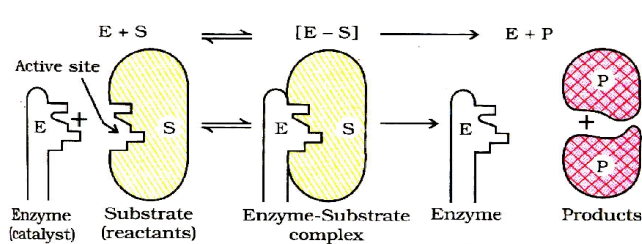
3.4 Enzyme catalysis

Enzymes are complex nitrogenous substances secreted by low forms of vegetables & organisms. Enzymes are actually protein molecules of higher molecular mass. (ranging from 15,000 – 1,000,000 g/mol) Enzymes form colloidal solutions in water and are very effective catalysts. They catalyse numerous reactions, especially those connected with natural processes. Numerous reactions occur in the bodies of animals and plants to maintain the life process. These reactions are catalysed by enzymes. The enzymes are thus, termed as **bio-chemical catalysts** and the phenomenon is known as **bio-chemical catalysis**. eg.

- Nitrogenase** is an enzyme present in bacteria on the root nodules of leguminous plants such as peas and beans, catalyses the conversion of atmospheric N_2 to NH_3 .
- In the human body, the enzyme **carbonic anhydrase** catalyses the reaction of CO_2 with H_2O ,

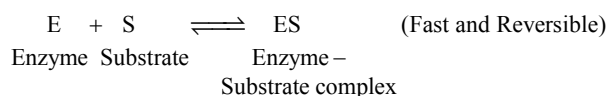


The forward reaction occurs when the blood takes up CO_2 in the tissues, and the reverse reaction occurs when the blood releases CO_2 in lungs.

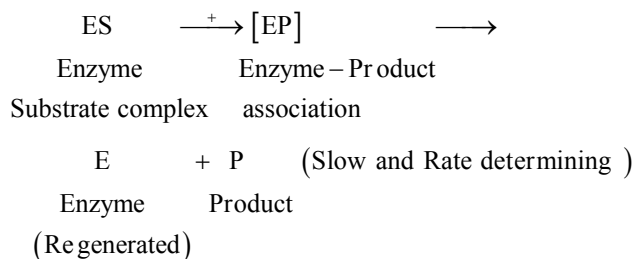


Thus, the enzyme catalysed reactions take place in two steps as follows :

Step 1. Formation of enzyme-substrate complex.



Step 2. Dissociation of enzyme-substrate complex to form the products.



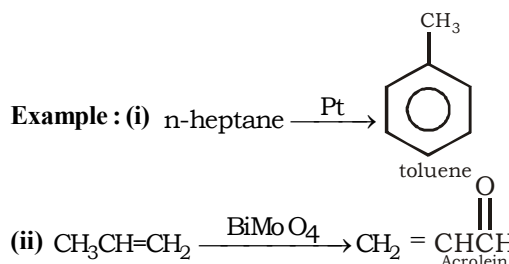
The rate of formation of product depends upon the concentration of ES.

3.5 Activity and selectivity

3.5.1 Activity : Activity is the ability of catalysts to accelerate chemical reactions. The degree of acceleration can be as high as 10^{10} times in certain reactions. For example reaction between H_2 and O_2 to form H_2O in presence of platinum as catalyst takes place with explosive violence.

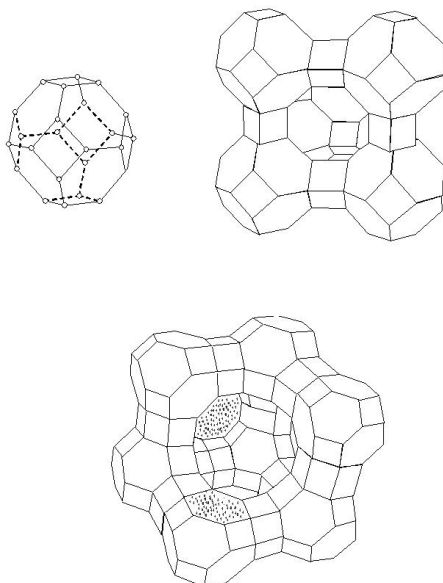
In absence of catalyst, H_2 and O_2 can be stored indefinitely without any reaction.

3.5.2 Selectivity : Is the ability of catalysts to direct reaction to yield particular products (excluding other).

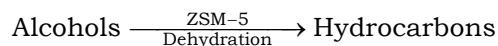


3.6 Zeolites (shape selective catalysis)

Zeolites are aluminosilicates of the general formula, $M_n [AlO_2]_x (SiO_2)_y \cdot mH_2O$, where, M may be simple cation like Na^+ , K^+ or Ca^{2+} , n is the charge on the simple cation, m is the number of molecules of water of crystallization. They have honey comb like structure called β cage. The reactions catalyzed by zeolites depend upon the shape and size of the reactant and product molecules. That is why these types of reactions are called shape selective-catalysis.



Zeolites are being very widely used as catalysts in petrochemical industries for cracking of hydrocarbons and isomerization. An important zeolite catalyst used in the petroleum industry is ZSM-5*** (Zeolite Sieve of Molecular Porosity 5). It converts alcohols directly into gasoline (petrol) by first dehydrating them so that a mixture of hydrocarbons is formed.



hydrated zeolites are used as 'ion-exchangers' in softening of hard water.

3.7 Catalysis used in Industries

SOME INDUSTRIAL CATALYTIC PROCESSES

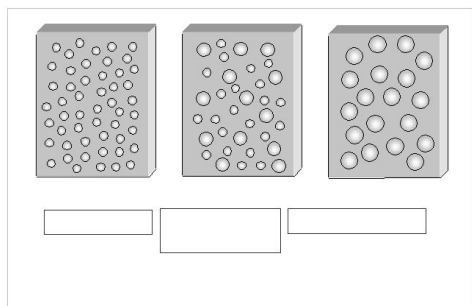
Process	Reactions with catalyst and other conditions
Haber's process for the manufacture of ammonia	$\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \xrightarrow[\substack{\text{Molybdenum (Promoter)} \\ 200 \text{ bar pressure} \\ 723-773 \text{ K temp.}}]{\text{Finely divided iron (catalyst)}} 2 \text{NH}_3(\text{g})$
Ostwald's process for the manufacture of nitric acid	$4 \text{NH}_3(\text{g}) + 5 \text{O}_2(\text{g}) \xrightarrow[\text{asbestos (catalyst)}]{\text{Platinised}} 4 \text{NO}(\text{g}) + 6 \text{H}_2\text{O}(\text{g})$ $2 \text{NO}(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{573\text{K}} 2 \text{NO}_2(\text{g})$ $4 \text{NO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) \rightarrow 4 \text{HNO}_3(\text{l})$
Contact process for the manufacture of sulphuric acid	$2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \xrightleftharpoons[\text{or } \text{V}_2\text{O}_5(\text{catalyst})]{\text{Platinised asbestos}} 2 \text{SO}_3(\text{g})$ $\text{SO}_3(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \xrightarrow{673-723\text{K}} \text{H}_2\text{S}_2\text{O}_7(\text{l}) \xrightarrow{+\text{H}_2\text{O}(\text{l})} 2 \text{H}_2\text{SO}_4(\text{aq})$ Oleum
Bosch's process for the manufacture of hydrogen.	$\underbrace{\text{CO} + \text{H}_2}_{\text{Water gas}} + \text{H}_2\text{O}(\text{g}) \xrightarrow[\substack{+\text{Cr}_2\text{O}_3(\text{promoter}) \\ 673-873\text{K}}]{\text{Fe}_2\text{O}_3(\text{Catalyst})} \text{CO}_2(\text{g}) + 2 \text{H}_2(\text{g})$
Deacon's process for the manufacture of chlorine	$4 \text{HCl}(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{773\text{K}} 2 \text{H}_2\text{O}(\text{l}) + 2 \text{Cl}_2(\text{g})$
Synthesis of methanol	$\text{CO}(\text{g}) + 2 \text{H}_2(\text{g}) \xrightarrow[432 \text{ K}]{\substack{\text{ZnO} + \text{Cr}_2\text{O}_3 \\ 200 \text{ bar}}} \text{CH}_3\text{OH}(\text{l})$
Hydrogenation of vegetable oils	$\text{Oil}(\text{l}) + \text{H}_2(\text{g}) \xrightarrow[\text{High pressure}]{\substack{\text{Finely divided Ni} \\ 423-473 \text{ K}}} \text{Vanaspati ghee}(\text{s})$

4. COLLOIDAL STATE

The colloidal state depends on the particle size. It is regarded as intermediate state between true solution and suspension.

4.1 True solution

In true solutions the size of the particles of solute is very small and thus, these can not be detected by any optical means and freely diffuse through membranes. It is a homogenous system.

**4.2 Suspension**

The size of particles is large and, thus it can be seen by naked eye and do not pass through filter paper. It is a heterogeneous system.

The size of different solutions are sometimes expressed in other units also as given below :

True solutions	Colloids	Suspensions	Relation
$<10^{-9}\text{m}$	10^{-9}m to 10^{-7}m	$>10^{-7}\text{m}$	
$<1\text{nm}$	$1\text{nm} - 100\text{nm}$	$>100\text{nm}$	$1\text{nm} = 10^{-9}\text{m}$
$<10\text{\AA}$	$10\text{\AA} - 1000\text{\AA}$	$>1000\text{\AA}$	$1\text{\AA} = 10^{-10}\text{m}$
$<1000\text{pm}$	$1000\text{pm} - 10^5\text{pm}$	$>10^5\text{pm}$	$1\text{pm} = 10^{-12}\text{m}$

CHARACTERISTICS OF TRUE SOLUTIONS, COLLOIDAL SOLUTIONS AND SUSPENSIONS

S.No	Property	True Solutions	Colloidal Solutions	Suspensions
1.	Nature	Homogeneous	Heterogeneous	Heterogeneous
2.	Particle size (diameters)	Less than 10^{-9}m or 1nm (i.e., $<10\text{\AA}$)	Between 10^{-9} to 10^{-6}m or 1nm to 1000nm	More than 10^{-6}m or 1000nm (i.e., $>10000\text{\AA}$)
3.	Filtrability	Pass through ordinary filter paper as well as animal membrane.	Pass through ordinary filter paper but not through animal membrane.	Do not pass through filter paper and animal membrane.
4.	Settling	Do not settle.	Do not settle.	Settle on standing.
5.	Visibility	Particles are invisible.	Scattering of light by the particles is observed under ultra-microscope.	Particles are visible to naked eye or under a microscope.
6.	Diffusion	Diffuse quickly.	Diffuse slowly.	Do not diffuse.
7.	Appearance	Clear and transparent	Translucent.	Opaque.

4.3 Dispersed phase and Dispersion Medium

4.3.1 Dispersed phase : (Discontinuous phase) : It is the component present in small proportion and is just like a solute in a solution. For example in the colloidal solution of silver in water (silver acts as a dispersed phase)

4.3.2 Dispersion medium : (continuous phase) : It is generally the component present in excess and is just like a

solvent in a solution. For example, in the colloidal solution of silver in water. Water acts as a dispersion medium.

4.4 Classification of Colloids

4.4.1 Classification based on the physical state of the dispersed phase and dispersion medium Depending upon the physical state of dispersed phase and dispersion medium whether these are solids, liquids or gases, eight types of colloidal systems are possible.

DIFFERENT TYPES OF COLLOIDAL SYSTEMS

Sr. No	Dispersed phase	Dispersion	Colloidal System	Examples
1.	Liquid	Gas	Aerosol of liquids	Fogs, clouds, mists, fine insecticide sprays
2.	Solid	Gas	Aerosol of solids	Smoke, volcanic dust, haze Foam or froth Soap lather.
3.	Gas	Liquid		Lemonade froth, foam, whipped cream, soda water
4.	Liquid	Liquid	Emulsions	Milk, emulsified oils, medicines
5.	Solid	Liquid	Sols	Most paints, starch in water, proteins, gold sol, arsenic sulphide sol, ink
6.	Gas	Solid	Solid foam	Pumice stone, styrene rubber, foam rubber
7.	Liquid	Solid	Gels	Cheese, butter, boot polish, jelly, curd
8.	Solid	Solid	Solid sols (coloured glass)	Ruby glass, some gem stones and alloys

4.4.2 Classification based on Nature of interaction between dispersed phase and dispersion medium :

- ♦ **Lyophilic colloids (water loving) :** “The colloidal solutions in which the particles of the dispersed phase have a great affinity for the dispersion medium, are called **lyophilic colloids.**”
- ♦ **Lyophobic colloids (water hating) :** “The colloidal solutions in which there is no affinity between particles of the dispersed phase and the dispersion medium are called **lyophobic colloids.**”
- ♦ **Lyophobic colloids (water hating) :** “The colloidal solutions in which there is no affinity between particles of the dispersed phase and the dispersion medium are called **lyophobic colloids.**”

DISTINCTION BETWEEN LYOPHILIC AND LYOPHOBIC SOLS

Sr. No	Property	Lyophilic (suspension)	Lyophobic Sols (Emulsion)
1.	Surface tension	Lower than that of the medium	Same as that of the medium
2.	Viscosity	Much higher than that of the medium	Same as that of the medium
3.	Reversibility	Reversible	Irreversible
4.	Stability	More stable	Less stable
5.	Visibility	Particles can't be detected even under ultramicroscope	Particles can be detected under ultramicroscope.
6.	Migration	Particles may migrate in either direction or do not migrate in an electric field because do not carry any charge.	Particles migrate either towards cathode or anode in an electric field because they carry charge.
7.	Action of electrolyte	Addition of smaller quantity of electrolyte has little effect	Coagulation takes place
8.	Hydration Examples	Extensive hydration takes place Gum, gelatin, starch, proteins, rubber etc. like As_2S_3 etc.	No hydration Metals like Ag and Au, hydroxides like $Al(OH)_3$, $Fe(OH)_3$, metal sulphides

4.4.3 Classification based on types of particles of dispersed phase : Depending upon the type of the particles of the dispersed phase, the colloids are classified as follows.

- ◆ **Multimolecular colloids :** When on dissolution, atoms or smaller molecules of substances (having diameter less than 1 nm) aggregate together to form particles of colloidal dimensions, the particles thus formed are called **multimolecular colloids**. In these sols the dispersed phase consists of aggregates of atoms or molecules with molecular size less than 1 nm . For example, sols of gold atoms and sulphur (S_8) molecules. In these colloids, the particles are held together by **Vander Waal's forces**. They have usually **lyophilic character**.
- ◆ **Macromolecular colloids :** These are the substances having big size molecules (called macromolecules) which on dissolution form size in the colloidal range. Such substances are called **macromolecular colloids**.

These macromolecules forming the dispersed phase are generally **polymers** having very high molecular masses. Naturally occurring macromolecules are starch, cellulose, proteins, enzymes, gelatin etc. Artificial macromolecules are synthetic polymers such as nylon, polythene, plastics, polystyrene etc. Their solutions are quite stable and resemble with true solution in many respects. They have usually **lyophobic character**. The molecules are flexible and can take any shape.

- ◆ **Associated colloids :** These are the substances which dissolved in a medium behave as normal electrolytes at low concentration but behave, as colloidal particles at higher concentration due to the formation of aggregated particles. The aggregates particles thus formed are called **micelles**. Their molecules contain both **lyophilic** and **lyophobic** groups. The colloidal behaviour of such substances is due to the formation of aggregates or clusters in solutions. Such aggregated particles are called **micelles**.

DIFFERENCE BETWEEN DIFFERENT TYPES OF COLLOIDS

Multimolecular colloids	Macromolecular colloids	Associated colloids
1. They are formed by the aggregation of a large number of atoms or molecules which generally have diameters less than 1 nm , e.g., sols of gold, sulphur, etc.	They are molecules of large size, e.g., polymers like rubber, nylon, starch, proteins, etc.	They are formed by of aggregation of a large no. of ions in concentrated solution e.g., soap sol.
2. Their molecular masses are not very high	They have high molecular masses.	Their molecular masses are generally high.
3. Their atoms or molecules are held together by weak van der Waals forces.	Due to long chain, the van der Waals forces holding them are comparatively stronger.	Higher is the concentration, greater are the van der Waals forces.

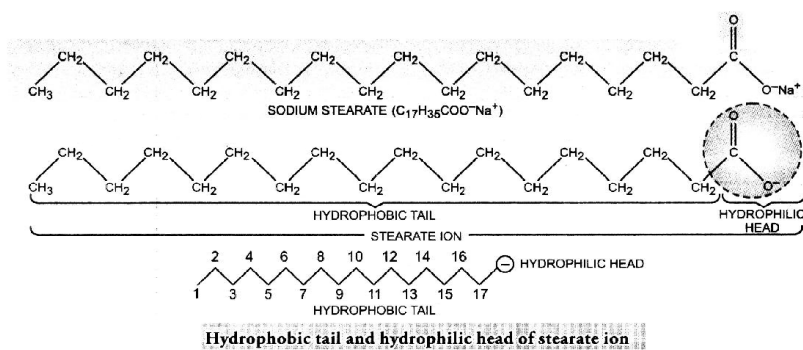
4.5 Micelles

Micelles are the clusters or aggregated particles formed by association of colloid in solution. The common examples of micelles are **soaps** and **detergents**. The formation of micelles takes place above a particular temperature called **Kraft temperature** (T_k) and above a particular concentration called **critical micellization concentration** (CMC). They are capable of forming ions. Micelles may contain as many as 100 molecules or more.

- ◆ For example sodium stearate ($C_{17}H_{35}COONa$) is a typical example of such type of molecules.
- ◆ When sodium stearate is dissolved in water, it gives Na^+ and $C_{17}H_{35}COO^-$ ions.

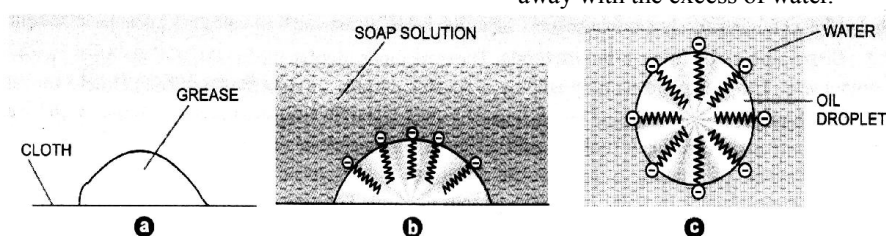
$$C_{17}H_{35}COONa \rightleftharpoons C_{17}H_{35}COO^- + Na^+$$

Sodium stearate
Stearate ion
- ◆ The stearate ions associate to form ionic micelles of colloidal size.
- ◆ It has long hydrocarbon part of $C_{17}H_{35}$ radical. Which is **lyophobic** and COO^- part which is **lyophilic**.



4.5.1 Cleansing action of Soaps : When oil droplet comes in contact with soap solution, the stearate ions arrange themselves around it in such a way that hydrophobic parts of the stearate ions are in the oil (or grease) and the hydrophilic.

As hydrophilic part is polar, these polar groups can interact with the water molecules present around the oil droplet. As a result, the oil droplet is pulled away from the surface of the cloth into water to form ionic micelle which is then washed away with the excess of water.



(a) Grease or oil on surface of cloth (b) Stearate ions arranged around the oil droplet
(c) Ionic micelle formed surrounded by sheath of negative charge.

In fact, the stearate ions of soap molecules help in making a stable emulsion of oil with water which is washed away with the excess of water. It may be noted that a sheath of negative charge is formed around the oil globule.

Similarly, in case of detergents, e.g., sodium lauryl sulphate, viz, $CH_3(CH_2)_{11}OSO_3^-Na^+$, the polar group is OSO_3^- alongwith the long hydrocarbon chain. It is an example of an **anionic** detergent as anions associate together to form an ionic micelle, similar to that of soap. A well known example of a cationic detergent forming an associated colloid is that of cetyl trimethyl ammonium bromide, $CH_3(CH_2)_{15}(CH_3)_3N^+Br^-$.

4.6 General Methods of preparation of Colloids

Lyophilic and lyophobic colloidal solutions (or sols) are generally prepared by different types of methods. Some of the common methods are as follows.

4.6.1 Preparation of Lyophilic colloids :

- The lyophilic colloids have strong affinity between particles of dispersed phase and dispersion medium. These colloidal

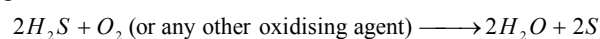
solutions are readily formed by simply mixing the dispersed phase and dispersion medium under ordinary conditions.

- For example, the substance like gelatin, gum, starch, egg, albumin etc. pass readily into water to give colloidal solution.

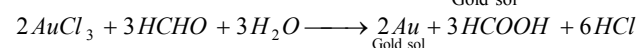
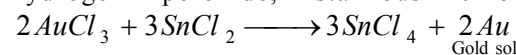
They are reversible in nature because these can be precipitated and directly converted into colloidal state.

4.6.2 Preparation of Lyophobic colloids :

4.6.2.1 By oxidation : A colloidal solution of sulphur can be obtained by bubbling oxygen (or any other oxidising agent like HNO_3 , Br_2 etc.) through a solution of hydrogen sulphide in water.



4.6.2.2 By reduction : A number of metals such as silver, gold and platinum, have been obtained in colloidal state by treating the aqueous solution of their salts, with a suitable reducing agent such as formaldehyde, phenyl hydrazine, hydrogen peroxide, stannous chloride etc.

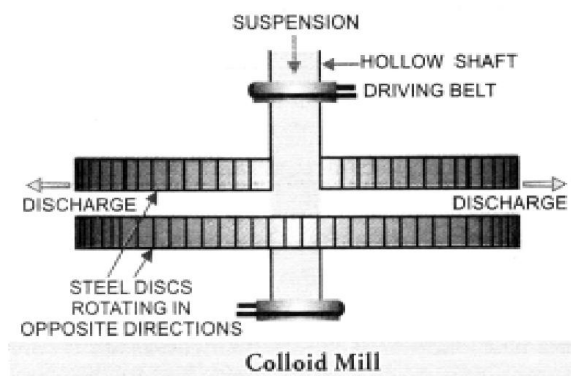


The gold sol, thus prepared, has a purple colour and is called purple of cassius.

4.6.2.3 By hydrolysis : Many salt solutions are rapidly hydrolysed by boiling dilute solutions of their salts. For example, ferric hydroxide and aluminium hydroxide sols are obtained by boiling solutions of the corresponding chlorides. $FeCl_3 + 3H_2O \longrightarrow Fe(OH)_3 + 3HCl$
Colloidal sol
Similarly silicic acid sol is obtained by the hydrolysis of sodium silicate.

4.6.2.4 By double decomposition : A sol of arsenic sulphide is obtained by passing hydrogen sulphide through a cold solution of arsenious oxide in water.
 $As_2O_3 + 3H_2S \longrightarrow As_2S_3 + 3H_2O$

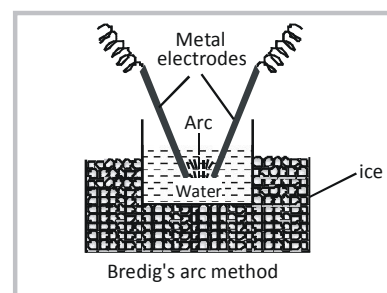
4.6.2.5 Mechanical dispersion : In this method, the substance is first ground to coarse particles. It is then mixed with the dispersion medium to get a suspension. The suspension is then grinded in colloidal mill. It consists of two metallic discs nearly touching each other and rotating in opposite directions at a very high speed about 7000 revolution per minute. The space between the discs of the mill is so adjusted that coarse suspension is subjected to great shearing force giving rise to particles of colloidal size. Colloidal solutions of black ink, paints, varnishes, dyes etc. are obtained by this method.



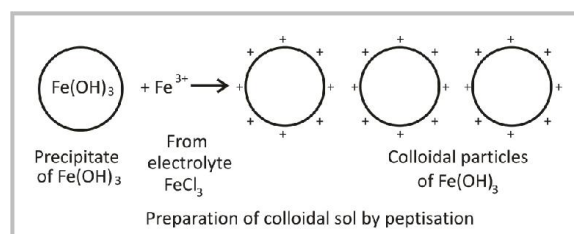
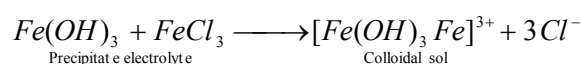
4.6.2.6 By electrical dispersion or Bredig's arc method : This method is used to prepare sols of platinum, silver, copper or gold.

The metal whose sol is to be prepared is made as two electrodes which are immersed in dispersion medium such as water etc. The dispersion medium is kept cooled by ice. An electric arc is struck between the electrodes. The

tremendous heat generates by this method and give colloidal solution. The colloidal solution prepared is stabilised by adding a small amount of KOH to it.



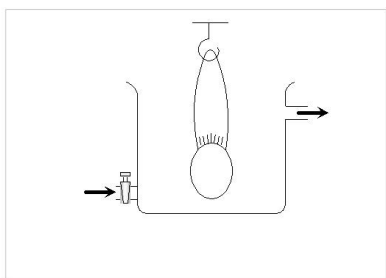
4.6.2.7 By Peptization : The process of converting a freshly prepared precipitate into colloidal form by the addition of suitable electrolyte is called **peptization**. The electrolyte is used for this purpose is called **peptizing agent** or **stabilizing agent**. Cause of peptisation is the adsorption of the ions of the electrolyte by the particles of the precipitate. Important peptizing agents are sugar, gum, gelatin and electrolytes. Freshly prepared ferric hydroxide can be converted into colloidal state by shaking it with water containing Fe^{3+} or OH^- ions, viz. $FeCl_3$ or NH_4OH respectively.



4.7 Purification of colloidal solution

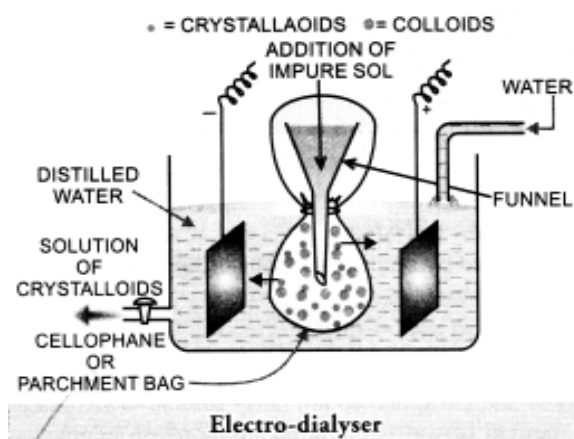
The colloidal solutions prepared by the above methods usually contain impurities especially electrolytes which can destabilize the sols. These impurities must be eliminated to make the colloidal solutions stable. The following methods are commonly used for the purification of colloidal solutions.

4.7.1 Dialysis : The process of separating the particles of colloid from those of crystalloid, by means of diffusion through a suitable membrane is called dialysis. Its principle is based upon the fact that colloidal particles can not pass through a parchment or cellophane membrane while the ions of the electrolyte can pass through it.



The colloidal solution is taken in a bag (parchment paper). The bag is suspended in fresh water. The impurities slowly diffused out of the bag leaving behind pure colloidal solution. The distilled water is changed frequently to avoid accumulation of the crystalloids otherwise they may start diffusing back into the bag. Dialysis can be used for removing HCl from the ferric hydroxide sol.

4.7.2 Electrodialysis : The ordinary process of dialysis is slow. To increase the pace of purification, the dialysis is carried out by applying electric field. This process is called **electrodialysis**. Kidneys in the human body act as dialysers to purify blood which is colloidal in nature. The important application of dialysis process in the artificial kidney machine used for the purification of blood of the patients whose kidneys have failed to work. The artificial kidney machine works on the principle of **dialysis**.



4.7.3 Ultra – filtration : Sol particles directly pass through ordinary filter paper because their pores are larger (more than 1μ or $1000m\mu$) than the size of sol particles (less than $200m\mu$). If the pores of the ordinary filter paper are made smaller by soaking the filter paper in a solution of gelatin or colloidion and subsequently hardened by soaking in

formaldehyde, the treated filter paper may retain colloidal particles and allow the true solution particles to escape. Such filter paper is known as **ultra – filter** and the process of separating colloids by using ultra – filters is known as **ultra – filtration**.

4.7.4 Ultra – centrifugation : The sol particles are prevented from setting out under the action of gravity by kinetic impacts of the molecules of the medium. The setting force can be enhanced by using high speed centrifugal machines having 15,000 or more revolutions per minute. Such machines are known as **ultra – centrifuges**.

4.8 Properties of colloidal solutions

The main characteristic properties of colloidal solutions are as follows.

4.8.1 Physical properties :

- ♦ **Heterogeneous nature :** Colloidal sols are heterogeneous in nature. They consist of two phases; the Dispersed Phase and the Dispersion Medium.
- ♦ **Stable nature :** The colloidal solutions are quite stable. Their particles are in a state of motion and do not settle down at the bottom of the container.
- ♦ **Filterability :** Colloidal particles are readily passed through the ordinary filter papers. However they can be retained by special filters known as ultrafilters (parchment paper).

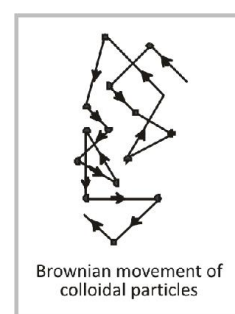
4.8.2 Colligative properties :

- ♦ Due to formation of associated molecules, observed values of colligative properties like relative decrease in vapour pressure, elevation in boiling point, depression in freezing point, osmotic pressure are smaller than expected.
- ♦ For a given colloidal sol the number of particles will be very small as compared to the true solution.

4.8.3 Mechanical properties :

- ♦ **Brownian movement :**

The colloidal particles are moving at random in a **zig – zag** motion. This type of motion is called **Brownian movement**.

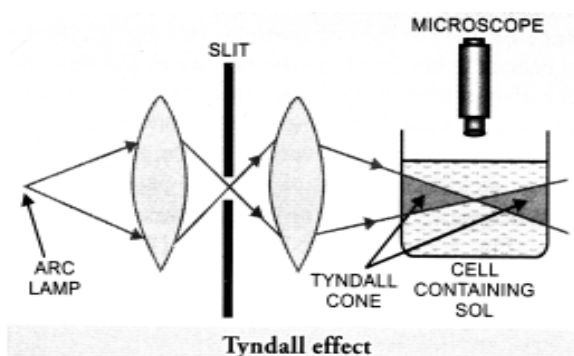


The reason of brownian motion is that the molecules of the dispersion medium are constantly colliding with the particles of the dispersed phase.

- ♦ **Diffusion** : The sol particles diffuse from higher concentration to lower concentration region. However, due to bigger size, they diffuse at a lesser speed.
- ♦ **Sedimentation** : *The colloidal particles settle down under the influence of gravity at a very slow rate. This phenomenon is used for determining the molecular mass of the macromolecules.*

4.8.4 Optical properties : Tyndall effect :

When light passes through a sol, its path becomes visible because of scattering of light by particles. It is called **Tyndall effect**. This phenomenon was studied for the first time by **Tyndall**. The illuminated path of the beam is called **Tyndall cone**. *In a true solution, there are no particles of sufficiently large diameter to scatter light and hence no Tyndall effect is observed. The Tyndall effect has also been observed by an instrument called **ultra – microscope**.*



4.8.5 Electrical properties :

Colloidal particles carry an electric charge and the dispersion medium has an opposite and equal charge, the system as a whole being electrically neutral. The presence of equal and similar charges on colloidal particles is largely responsible in giving stability to the system because the mutual forces of repulsion between similarly charged particles prevent them from coalescing and coagulating when they come closer to one another.

4.9 Origin of the charge on colloidal particles

The origin of the charge on the sol particles in most cases is due to the preferential adsorption of either positive or

negative ions on their surface. The sol particles acquire electrical charge in any one or more of the following ways.

4.9.1 Due to the dissociation of the surface molecules:

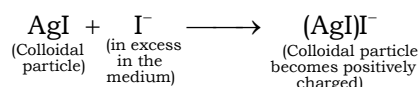
Some colloidal particles develop electrical charge due to the dissociation / ionisation of the surface molecules. The charge on the colloidal particles is balanced by the oppositely charged ions in the sol. For example, an aqueous solution of soap (sodium palmitate) which dissociates into ions as, $C_{15}H_{31}COONa \rightleftharpoons C_{15}H_{31}COO^- + Na^+$.

Sodium palmitate

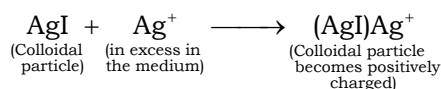
The cations (Na^+) pass into the solution while the anions ($C_{15}H_{31}COO^-$) have a tendency to form aggregates due to weak attractive forces present in the hydrocarbon chains.

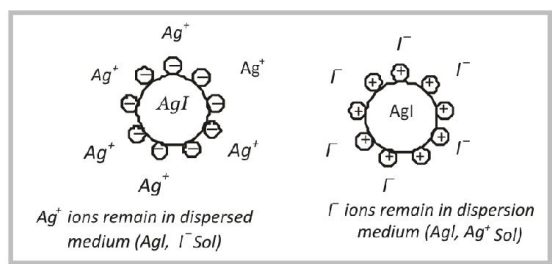
4.9.2 Due to frictional electrification : It is believed that the frictional electrification due to the rubbing of the dispersed phase particles with that of dispersion medium results in some charge on the colloidal particles. The dispersion medium must also get some charge, because of the friction. Since it does not carry any charge, the theory does not seem to be correct.

4.9.3 Due to selective adsorption of ions : The particles constituting the dispersed phase adsorb only those ions preferentially which are common with their own lattice ions. For example, when a small quantity of silver nitrate ($AgNO_3$) solution is added to a large quantity of potassium iodide (KI) solution, the colloidal particles of silver iodide adsorb I^- from the solution to become **negatively charged**, (at this stage KI is in excess, and I^- being common to AgI)

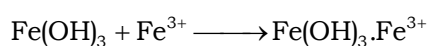


But, when a small quantity of potassium iodide (KI) solution is added to a large quantity of silver nitrate solution ($AgNO_3$) ; the colloidal silver iodide particles adsorb Ag^+ from the solution to become **positively charged**, (at this stage $AgNO_3$ is in excess and Ag^+ is common to AgI),



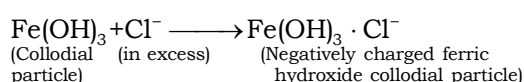
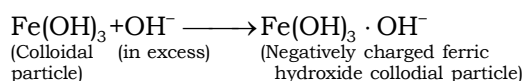


Similarly, the ferric hydroxide colloidal particles develop positive charge due to the adsorption of Fe^{3+} ions from the solution.



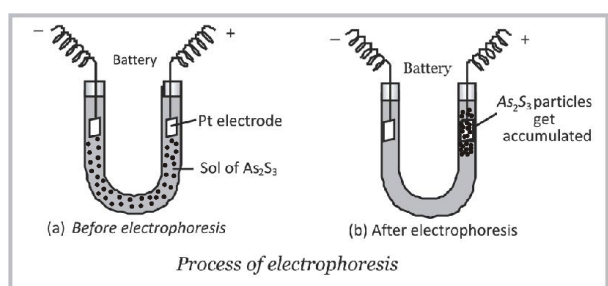
(Colloidal particle) (in excess in the medium) (Colloidal particle of ferric hydroxide get positively charged due to the adsorption of Fe^{3+} ions on it)

Ferric hydroxide colloidal particles develop negative charge due to adsorption of either OH^- or



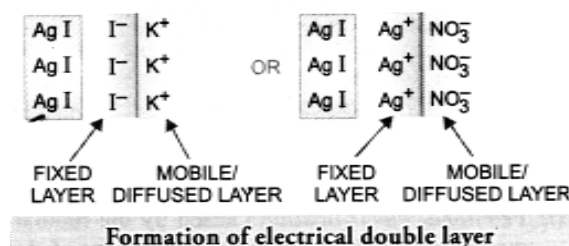
4.10 Electrophoresis

The phenomenon of movement of colloidal particles under an applied electric field is called **electrophoresis**. If the particles accumulate near the negative electrode, the charge on the particles is **positive**. On the other hand, if the sol particles accumulate near the positive electrode, the charge on the particles is **negative**.

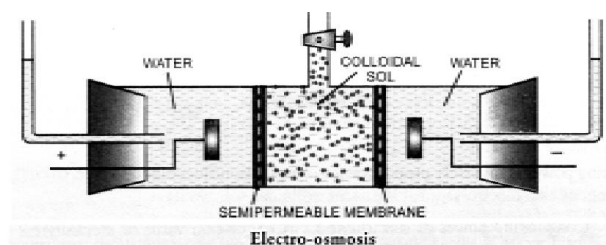


4.10.1 Electrical double layer theory : The electrical properties of colloids can also be explained by electrical double layer theory. According to this theory **a double layer of ions appear at the surface of solid**. The ion preferentially adsorbed is held in fixed part and imparts charge to colloidal particles. The second part consists of a diffused mobile layer

of ions. This second layer consists of both the type of charges. The net charge on the second layer is exactly equal to that on the fixed part. The existence of opposite sign on fixed and diffuse parts of double layer leads to appearance of a difference of potential, known as **zeta potential or electrokinetic potential**. Now when electric field is employed the particles move (electrophoresis)



4.10.2 Electro-osmosis : In it the movement of the dispersed particles is prevented from moving by **semipermeable membrane**. Electro-osmosis is a phenomenon in which dispersion medium is allowed to move under the influence of an electrical field, whereas colloidal particles are not allowed to move. The existence of electro-osmosis has suggested that when liquid is forced through a porous material or a capillary tube, a potential difference is setup between the two sides called as **streaming potential**.



4.11 Stability of sols

Sols are thermodynamically unstable and the dispersed phase (colloidal particles) tend to separate out on long standing due to **Vander Waal's attractive forces**. However sols tend to exhibit some stability due to

4.11.1 Stronger repulsive forces between the similarly charged particles : All colloidal particles in any sol possess similar charge. Therefore, due to the electrostatic repulsion these are not able to come closer and form aggregates. Thus stronger repulsive forces between the similarly charged particles in a sol promote its stability.

4.11.2 Particle-solvent interactions : Due to strong particle-solvent (dispersion medium) interactions, the colloidal particles get strongly solvated. Due to solvation, the effective distance between the colloidal particles increases, and therefore, the Vander Waal's force of attraction decreases. As a result, the particles are not able to form aggregates. Lyophilic sols are mainly stabilized by solvation effects due to strong interactions between the sol particles and the dispersion medium.

4.12 Coagulation or Flocculation or Precipitation

*“The phenomenon of the precipitation of a colloidal solution by the addition of the excess of an electrolyte is called **coagulation or flocculation**.”*

4.12.1 By electrophoresis : In electrophoresis the colloidal particles move towards oppositely charged electrode. When these come in contact with the electrode for long these are discharged and precipitated.

4.12.2 By mixing two oppositely charged sols : When oppositely charged sols are mixed in almost equal proportions, their charges are neutralised. Both sols may be partially or completely precipitated as the mixing of ferric hydroxide (+ve sol) and arsenious sulphide (–ve sol) brings them in precipitated form. This type of coagulation is called **mutual coagulation or metereal coagulation**.

4.12.3 By boiling : When a sol is boiled, the adsorbed layer is disturbed due to increased collisions with the molecules of dispersion medium. This reduces the charge on the particles and ultimately they settle down to form a precipitate.

4.12.4 By persistent dialysis : On prolonged dialysis, the traces of the electrolyte present in the sol are removed almost completely and the colloids become unstable.

4.12.5 By addition of electrolytes : The particles of the dispersed phase *i.e.*, colloids bear some charge. When an electrolyte is added to sol, the colloidal particles take up ions carrying opposite charge from the electrolyte. As a

result, their charge gets neutralised and this causes the uncharged, particles to come closer and to get coagulated or precipitated. For example, if $BaCl_2$ solution is added to As_2S_3 sol the Ba^{2+} ions are attracted by the negatively charged sol particles and their charge gets neutralised. This leads to coagulation.

4.12.6 Hardy schulze rule :

The coagulation capacity of different electrolytes is different. It depends upon the valency of the active ion are called flocculating ion, which is the ion carrying charge opposite to the charge on the colloidal particles. *Greater the valency of the active ion or flocculating ion, greater will be its coagulating power”.*

Coagulating power of an electrolyte is directly proportional to the valency of the active ions (ions causing coagulation).

For example to coagulate **negative sol** of As_2S_3 , the **coagulation power** of different cations has been found to **decrease** in the order as, $Al^{3+} > Mg^{2+} > Na^+$

Similarly, to coagulate a **positive sol** such as $Fe(OH)_3$, the **coagulating power** of different anions has been found to **decrease** in the order :



4.12.7 Coagulation or flocculation value :

*“The minimum concentration of an electrolyte which is required to cause the coagulation or flocculation of a sol is known as **flocculation value**.”*

or

*“The number of millimoles of an electrolyte required to bring about the coagulation of one litre of a colloidal solution is called its **flocculation value**.”*

Thus, a more efficient flocculating agent shall have lower flocculating value.

FLOCCULATION VALUES OF SOME ELECTROLYTES

Sol	Electrolyte value (mM)	Flocculation	Sol Electrolyte	Flocculation value (mM)	
As ₂ S ₃ (-vely charged)	NaCl	51.0	Fe(OH) ₃ (+vely charged)	KCl	9.5
	KCl	49.5		BaCl ₂	9.3
	CaCl ₂	0.65		K ₂ SO ₄	0.20
	MgCl ₂	0.72		MgSO ₄	0.22
	MgSO ₄	0.81			
	AlCl ₃	0.093			
	Al ₂ (SO ₄) ₃	0.096			
	Al(NO ₃) ₃	0.095			

Note : Coagulating value or flocculating value $\propto \frac{1}{\text{coagulating power}}$.

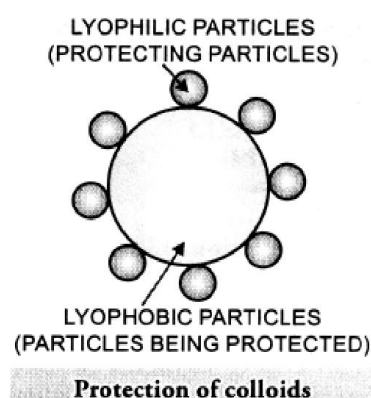
4.12.8 Coagulation of lyophilic sols : There are two factors which are responsible for the stability of lyophilic sols. These factors are the **charge** and **solvation of the colloidal particles**. When these two factors are removed, a lyophilic sol can be coagulated. This is done by **adding electrolyte** and by **adding suitable solvent**. When solvent such as alcohol and acetone are added to hydrophilic sols the dehydration of dispersed phase occurs. Under this condition a small quantity of electrolyte can bring about coagulation.

4.13 Protection of colloids and Gold number

Lyophilic sols are more stable than lyophobic sols. Lyophobic sols can be easily coagulated by the addition of small quantity of an electrolyte. When a lyophilic sol is added to any lyophobic sol, it becomes less sensitive towards electrolytes. Thus, lyophilic colloids can prevent the coagulation of any lyophobic sol.

“The phenomenon of preventing the coagulation of a lyophobic sol due to the addition of some lyophilic colloid is called sol protection or protection of colloids.” The protecting power of different protective (lyophilic) colloids is different. The efficiency of any protective colloid is expressed in terms of **gold number**.

4.13.1 Gold number : *Zsigmondy* introduced a term called **gold number** to describe the protective power of different colloids. This is defined as, “**weight of the dried protective agent in milligrams, which when added to 10 ml of a standard gold sol (0.0053 to 0.0058%) is just sufficient to prevent a colour change from red to blue on the addition of 1 ml of 10 % sodium chloride solution, is equal to the gold number of that protective colloid.**”



Thus, smaller is the gold number, higher is the protective action of the protective agent.

$$\text{Protective power} \propto \frac{1}{\text{Gold number}}$$

GOLD NUMBERS OF SOME HYDROPHILLIC SUBSTANCES

Sr. No.	Hydrophilic substance	Gold number	Hydrophilic substance	Gold number
1.	Gelatin	0.005-0.01	Sodium oleate	0.4 – 1.0
2.	Sodium caseinate	0.01	Gum tragacanth	2
3.	Haemoglobin	0.03-0.07	Potato starch	25
4.	Gum arabic	0.15-0.25		

The protective colloids play very significant role in stabilisation of the non-aqueous dispersions, such as paints, printing inks etc.

4.13.2 Congo rubin number : *Ostwald* introduced congo rubin number to account for protective nature of colloids. It is defined as “the amount of protective colloid in milligrams which prevents colour change in 100 ml of 0.01 % congo rubin dye to which 0.16 g equivalent of KCl is added.”

4.14 Application of colloids

4.14.1 Purification of water by alum (coagulation) : Alum which yield Al^{3+} ions, is added to water to coagulate the negatively charged clay particles.

4.14.2 In rubber and tanning industry (coagulation and mutual coagulation) : Several industrial processes such as rubber plating, chrome tanning, dyeing, lubrication etc are of colloidal nature.

In rubber plating, the negatively charged particles of rubber (latex) are made to deposit on the wires or handle of various tools by means of electrophoresis. The article on which rubber is to be deposited is made anode.

In tanning the positively charged colloidal particles of hides and leather are coagulated by impregnating them in negatively charged tanning materials (present in the barks of trees). Among the tanning agent chromium salts are most commonly used for the coagulation of the hide material and the process is called **chrome tanning**.

4.14.3 Artificial rains : It is possible to cause artificial rain by throwing the electrified sand or silver iodide from an aeroplane and thus coagulating the mist hanging in air.

4.14.4 Smoke precipitation (Coagulation) : Smoke is a *negative* sol consisting of carbon particles dispersed in air. Thus, these particles are removed by passing through a chamber provided with highly positively charged metallic knob.

4.14.5 Formation of deltas (Coagulation) : River water consists of negatively charged clay particles of colloidal dimension. When the river falls into the sea, the clay particles are coagulated by the positive Na^+ , K^+ , Mg^{2+} ions etc. present in sea water and new lands called deltas are formed.

4.14.6 Clotting of blood : Blood consists of negatively charged colloidal particles (albuminoid substance). The colloidal nature of blood explains why bleeding stops by applying a ferric chloride solution to the wound. Actually ferric chloride solution causes coagulation of blood to form a clot which stops further bleeding.

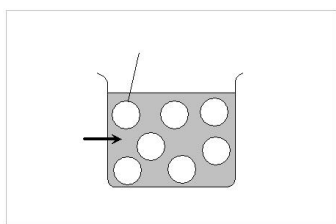
4.14.7 Colloidal medicine : Argyrol and protargyrol are colloidal solution of silver and are used as eye lotions. Colloidal sulphur is used as disinfectant. Colloidal gold, calcium and iron are used as tonics.

4.14.8 Photographic plates : These are thin glass plates coated with gelatin containing a fine suspension of silver bromide. The particles of silver bromide are colloidal in nature.

5. EMULSIONS

“Emulsions are the colloidal solutions in which both the dispersed phase and the dispersion medium are liquids.”

A good example of an emulsion is milk in which fat globules are dispersed in water. The size of the emulsified globules is generally of the order of 10^{-6} m. Emulsion resemble lyophobic sols in some properties.

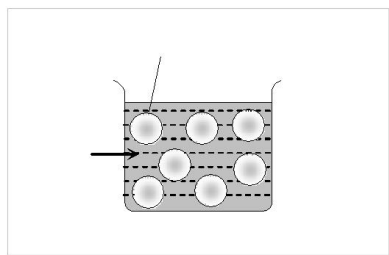


5.1 Types of Emulsion

Depending upon the nature of the dispersed phase, the emulsions are classified as;

phase, the emulsions are classified as;

5.1.1 Oil-in-water emulsions (O/W) : The emulsion in which oil is present as the dispersed phase and water as the dispersion medium (continuous phase) is called an oil-in-water emulsion. Milk is an example of the oil-in-water type of emulsion. In milk liquid fat globules are dispersed in water. Other examples are, vanishing cream etc.



5.1.2 Water-in-oil emulsion (W/O) : The emulsion in which water forms the dispersed phase, and the oil acts as the dispersion medium is called a water-in-oil emulsion. These emulsion are also termed **oil emulsions**. **Butter** and **cold cream** are typical examples of this types of emulsions. Other examples are cod liver oil etc.

5.2 Preparation of Emulsions

Emulsions are generally prepared by vigorously agitating a mixture of the relevant oil and water by using either a high speed mixer or by using ultrasonic vibrators. The emulsions obtained by simple mechanical stirring are unstable. The two components (oil and water) tend to separate out. To obtain a stable emulsion, a suitable stabilizing substance is generally added. The stabilizing substance is called **emulsifier or emulsifying agent**. The emulsifier is added

along with the oil and water in the beginning. For Examples : substances which can act as emulsifiers are soaps, detergents, long chain sulphonic acid, lyophilic colloids like gelatin, albumin, casein etc.

5.3 Nature of emulsifier

Different emulsifiers may act differently in the case of a particular emulsion.

For example,

- ♦ Sodium oleate is used to prepare oil-in-water (O/W) emulsions.
- ♦ Magnesium and calcium oleates are used to prepare water-in-oil (W/O) emulsions. When calcium oleate is added to an emulsion stabilized by sodium oleate, the stability of the system decreases. At a certain ratio of $Na^+ : Ca^{2+}$, the oil-in-water emulsion becomes unstable. If the Ca^{2+} ions concentration is increased further very quickly, then the reversal of the emulsion type occurs, that is the oil-in-water emulsion gets converted into a water-in-oil type.

5.4 Identification of emulsions

Several methods are available to find out whether an emulsion is of the oil-in-water type or of the water-in-oil type emulsion. An emulsion can be identified as follows.

5.4.1 Dilution test : Add water to the emulsion. If the emulsion can be diluted with water this means that water acts as the dispersion medium and it is an example of oil-in-water emulsion. In case it is not diluted, then oil acts as dispersion medium and it is an example of water-in-oil emulsion.

5.4.2 Dye test : An oil soluble suitable dye is shaken with the emulsion. If colour is noticed on looking at a drop of the emulsion, it is oil-in-water type emulsion. In case the entire background is coloured, it is an example of water-in-oil type.

5.4.3 Conductivity test : Add small amount of an electrolyte (e.g. KCl) to the emulsion. If this makes the emulsion electrically conducting, then water is the dispersion medium. If water is not the dispersed phase.

5.5 Properties of emulsion

5.5.1 Emulsions show all the characteristic properties of colloidal solution such as Brownian movement, Tyndall effect, electrophoresis etc.

5.5.2 These are coagulated by the addition of electrolytes containing polyvalent metal ions indicating the negative charge on the globules.

5.5.3 The size of the dispersed particles in emulsions is larger than those in the sols. It ranges from $1000 \text{ \AA}$ to $10,000 \text{ \AA}$. However, the size is smaller than the particles in suspensions.

5.5.4 Emulsions can be converted into two separate liquids by heating, centrifuging, freezing etc. This process is also known as **demulsification**.

5.6 Applications of emulsions

5.6.1 Concentration of ores in metallurgy

5.6.2 In medicine (Emulsion water-in-oil type)

5.6.3 Cleansing action of soaps.

5.6.4 Milk, which is an important constituent of our diet an emulsion of fat in water.

5.6.5 Digestion of fats in intestine is through **emulsification**.

SOLVED EXAMPLES**Example – 1**

Why is ferric chloride preferred over potassium chloride in case of a cut leading to bleeding ?

Sol. Fe^{3+} ion has greater coagulating power than Na^+ ion as it has higher charge.

Example – 2

What are the physical states of dispersed phase and dispersion medium of froth ?

Sol. Dispersed phase : Gas
Dispersion medium : Liquid

Example – 3

What causes Brownian movement in a colloidal solution ?

Sol. This is due to the unequal bombardment of colloidal particles by the molecules of dispersion medium.

Example – 4

Which has a higher enthalpy of adsorption, physisorption or chemisorption ?

Sol. Chemisorption

Example – 5

Mention two ways by which lyophilic colloids can be coagulated.

Sol. This can be done
(i) by adding an electrolyte
(ii) by adding a suitable solvent.

Example – 6

Define the term 'Tyndall effect'.

Sol. The scattering of light by colloidal particles is known as Tyndall effect.

Example – 7

What is the 'coagulation' process.

Sol. The process of settling of colloidal particles is called coagulation.

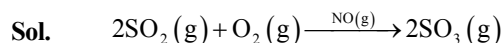
Example – 8

What is an emulsion?

Sol. Emulsion is a colloidal solution in which both the dispersed phase and dispersion medium are liquids e.g. milk, cod liver oil, etc.

Example – 9

Indicate a chemical reaction involving a homogeneous catalyst.


Example – 10

Give an example of a shape selective catalyst.

Sol. Zeolites are shape selective catalysts. A zeolite called ZSM-5 converts alcohol to gasoline.

Example – 11

In what way is a sol different from a gel ?

Sol. Colloidal system in which solid is dispersed in liquid is called sol and that in which liquid is dispersed in solid is called gel.

Example – 12

Why is a colloidal sol stable ?

Sol. All the particles in a colloidal sol carry the same charge and hence keep on repelling each other and cannot aggregate together to form bigger particles.

Example – 13

Write two differences between sols and emulsions.

Sol. (i) Sols are dispersions of solids in liquids while emulsions are dispersions of liquids in liquids.
(ii) Sols are quite stable whereas emulsions are less stable.

Example – 14

What is Kraft temperature ?

Sol. Kraft temperature is the minimum temperature above which the formulation of micelles takes place.

Example – 15

Give an example of an associated colloid.

Sol. Soaps and detergents are associated colloids.

Example – 16

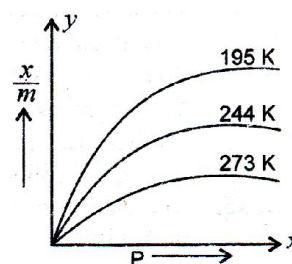
Define ultrafiltration.

Sol. In this process, colloidal solutions are purified by carrying out filtration through special types of graded filters called ultra-filters. Filter paper allows the passage of electrolyte but does not allow the passage of colloidal particles.

Example – 17

Consider the adsorption isotherms given below and interpret the variation in the extent of adsorption (x/m) when

- (i) temperature increases at constant pressure
 - (ii) pressure increases at constant temperature.
- (b) Name the catalyst and the promoter used in Haber's process for manufacture of ammonia.



Sol. (a) (i) Extent of adsorption ($\frac{x}{m}$) decreases with increase in temperature at constant pressure as adsorption is an exothermic process.
(ii) Extent of adsorption increases with increase in pressure at constant temperature. At low value of pressure.

$$\frac{x}{m} \propto P \quad \text{or} \quad \frac{x}{m} = KP$$

At high pressure

$$\frac{x}{m} \propto P^0 \quad \text{or} \quad \frac{x}{m} = KP^0 \quad \text{i.e.,} \quad \frac{x}{m} = K$$

In the intermediate range of pressure

$$\frac{x}{m} \propto P^{1/n}$$

or $\frac{x}{m} = KP^{1/n} \quad (n > 1)$

(b) Catalyst : Finely divided iron.

Promoter: Molybdenum.

Example – 18

Explain the following observations:

- (a) Ferric hydroxide sol gets coagulated on addition of sodium chloride solution.
- (b) Cottrell's smoke precipitator is fitted at the mouth of the chimney used in factories.
- (c) Physical adsorption is multilayered, while chemisorption is monolayered.

- Sol.**
- (a) As ferric hydroxide, $\text{Fe}(\text{OH})_3$ is a positively charged sol, so it gets coagulated by chloride ions Cl^- , released by NaCl solution.
 - (b) Cottrell's smoke precipitator, neutralises the charge on unburnt carbon particles, coming out of chimney and they get precipitated and settle down at the floor of the chamber.
 - (c) As physical adsorption, involves only weak van der waal's force of interaction, so many layers of adsorbate get attached, while chemisorption involves chemical bond formation between adsorbate and adsorbents monolayer is formed.

Example – 19

- (a) Adsorption of a gas on the surface of solid is generally accompanied by a decrease in entropy. Still it is a spontaneous process. Explain.
- (b) How does an increase in temperature affect both physical as well as chemical adsorption ?

- Sol.** (a) According to the equation

$$\Delta G = \Delta H - T\Delta S$$

for a process to be spontaneous ΔG should be negative. Even though ΔG is negative here, ΔG is negative because reaction is highly exothermic i.e., ΔH is negative.

(b) On increasing temperature desorption occurs in physical adsorption.

Chemical adsorption increases first and then decreases with increase in temperature.

Example – 20

What are lyophilic and lyophobic sols ? Give one example of each type. Which one of these two types of sols is easily coagulated and why?

Sol. **Lyophilic sols :** Lyophilic sols are those sols in which the particles of dispersed phase have great affinity for the dispersion medium, e.g., sols of gum, gelatine, starch, etc.

Lyophobic sols : In this type of sols the particles of dispersed phase have little or no affinity for the dispersion medium, e.g., gold sol, $\text{Fe}(\text{OH})_3$ sol, As_2O_3 sol., etc.

Example – 21

How are the following colloids different from each other in respect of dispersion medium and dispersed phase ? Give one example of each type.

- (i) An aerosol (ii) A hydrosol (iii) An emulsion

Sol.

Type of Colloid	Dispersed phase	Dispersion medium	Example
Aerosol	Solid or liquid	Gas	Smoke, Fog, Cloud
Hydrosol	Solid	Water	Starch sol, Protein sol
Emulsion	Liquid	Liquid	Milk, hair cream

Example – 22

What happens in the following activities and why?

- (i) An electrolyte is added to a hydrated ferric oxide sol in water.
- (ii) A beam of light is passed through a colloidal solution.
- (iii) An electric current is passed through a colloidal solution.

Sol. (i) The positively charged colloidal particles of $\text{Fe}(\text{OH})_3$ get coagulated by the negatively charged ions provided by electrolyte.

(ii) The path of light becomes visible due to scattering of light by colloidal particles (Tyndall effect).

(iii) Electrophoresis takes place in which colloidal particles move towards the oppositely charged electrode where they lose their charge and get coagulated.

Example – 23

How do size of particles of adsorbent, pressure of gas and prevailing temperature influence the extent of adsorption of a gas on a solid ?

Sol. The influence is in the following ways :

(i) Smaller the size of the particles of the adsorbent, greater is the surface area and greater is the adsorption.

(ii) At constant temperature, adsorption first increases with increase of pressure and then attains equilibrium at a high pressure.

(iii) In physical adsorption, it decreases with increase of temperature but in chemisorption, first it increases and then decreases.

Example – 24

What is meant by 'shape selective' catalysis?

Sol. Shape selective catalysis is a chemical reaction in which the rate depends on the pore size of the catalyst and also on the shape and size of the reactant and product molecules. Zeolite acts as a shape selective catalyst. For example, ZSM-5 (Zeolite sieve of molecular porosity 5) is used as a shape selective catalyst to produce gasoline of high octane number from methanol.

Example – 25

Write two differences between multimolecular colloids and macromolecular colloids.

Sol. (i) Multimolecular colloids are formed by aggregation of small molecules (diameter < 1 nm) while macromolecular colloids are formed by macromolecules (polymers) and consist of single molecules.

(ii) Multimolecular colloids are generally lyophobic whereas macromolecular colloids are generally lyophilic.

Example – 26

Explain the following observations :

- (a) Lyophilic colloid is more stable than lyophobic colloid.
- (b) Coagulation takes place when sodium chloride solution is added to a colloidal solution of ferric hydroxide.
- (c) Sky appears blue in colour.

Sol. (a) It is due to greater force of attraction between dispersed phase and dispersion medium in lyophilic colloid than lyophobic colloid.

(b) $\text{Fe}(\text{OH})_3$ sol is positively charged which is coagulated by negatively charged Cl^- present in Sodium Chloride solution.

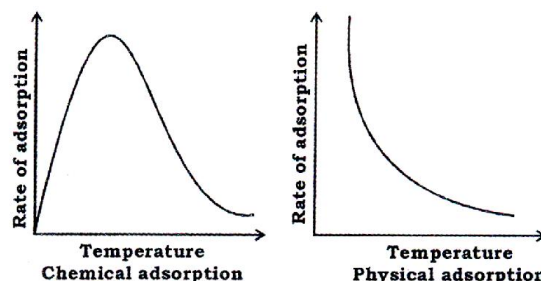
(c) Sky appears blue in colour due to scattering of light by colloidal particles. This is known as Tyndall effect.

Example – 27

Explain factors affecting adsorption.

Sol. Adsorption is generally affected by following factors:

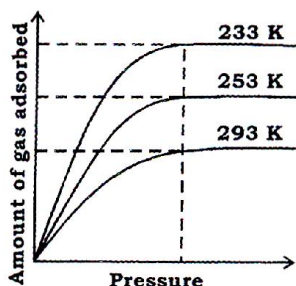
(i) Temperature : Adsorption is an exothermic process. Hence, according to Le Chatelier's principle, the rate of adsorption decreases with the increase in temperature and vice versa. Thus, adsorption is inversely proportional to temperature at a constant pressure. In the graphs plotted, showing the relationship between the rate of adsorption and the temperature at a constant pressure, the physical adsorption is favoured at low and chemical adsorption at relatively high temperature.



EFFECT OF TEMPERATURE ON RATE OF ADSORPTION

(ii) Pressure : The curve shows the effect of temperature on the rate of the adsorption at constant

pressure, is called Adsorption Isobar. In case of gases, adsorption increases with increase in the pressure of the gas. However, increase is not uniform. At low pressure the adsorption increases linearly while at moderate pressure it increases exponentially. At high pressure complete surface area of adsorbent is covered and hence adsorption becomes independent of pressure.



EFFECT OF PRESSURE ON ADSORPTION AT DIFFERENT TEMPERATURES

(iii) **Nature of adsorbent :** The gases which are easily liquified and highly soluble in water are easily adsorbed. NH_3 , HCl , Cl_2 , SO_2 , etc are adsorbed readily. These gases have greater Van der Waal's forces of attraction and hence get easily adsorbed. As molecular weight increases, adsorption increases. e.g. SO_2 is adsorbed more than CO_2 at the same temperature.

(iv) **Nature of Adsorbent :** Adsorption also depends on the nature of adsorbent. The adsorbent with more free valencies and unbalanced forces adsorb to greater extent. e.g. the transition metals. The same gas can be adsorbed to different extents on different adsorbents.

e.g. H_2 is adsorbed by different metals in the following order. $\text{Pd} > \text{Pt} > \text{Ni} > \text{Fe}$

(v) **Surface area of adsorbent :** Since, adsorption is the surface phenomenon, the rate of adsorption is directly proportional to the surface area of the adsorbent. Thus, greater the total surface area of the adsorbent, greater is the adsorption. Therefore, finely divided particles of the adsorbent show greater adsorption due to their large surface area.

(vi) **Concentration of adsorbate :** The extent of adsorption depends upon the concentration of the adsorbate, used in a solution. The rate of adsorption

increases with the increase in concentration of solution of the adsorbate, but upto certain limit. After the equilibrium is achieved, further increase in the concentration has no effect on the rate of adsorption.

Example – 28

Distinguish between Homogenous and Heterogenous catalysis.

Sol.

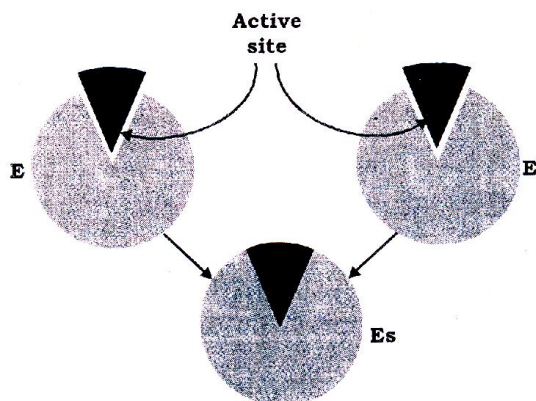
No.	Homogenous catalysis	No.	Heterogenesis catalysis
1.	The catalyst and the reactants form a single phase.	1.	The catalyst and the reactants are in different phases.
2.	The catalyst dissolves into the gas phase or solution (a reacting mixture).	2.	The catalyst does not dissolve in the reacting mixture.
3.	The reaction occurs in the liquid phase or gas phase.	3.	The reaction does not occur in gas or liquid phase but occurs on the surface of solid catalyst
4.	The catalyst is often involved in the chemical reaction.	4.	The catalyst adsorbs the reactant molecules to form a chemical bond between catalyst and the reactant molecules.
5.	The catalyst cannot be easily separated from the products of reaction.	5.	The catalyst can be easily separated from reaction products.
6.	The rate of the reaction does not depend on the surface area of catalyst	6.	The rate of reaction is proportional to the surface area of catalyst
7.	The homogeneously catalysed reactions are often little faster than those which are heterogeneously catalysed	7.	The heterogeneously catalysed reactions are often slower than those which are homogeneously catalysed

Example – 29

Give the mechanism of enzyme catalysis.

Sol.

An enzyme molecule has one or more active sites at which a specific substrate (reactant) molecule fits. In other words the active site acts like a lock into which fits only a specific key (substrate or reactant). This concept of lock and key was developed by German chemist Emil Fischer in 1894. Now it is known that the structure of active site is flexible and hence it can accommodate more than one type of substrate. The active site returns to its original state after the products are released.



The binding of substrate (reactant) at the active site of the enzyme catalyst produces enzyme – substrate complex which then decomposes to give the product of reaction and the enzyme is recovered.



where, E and S are enzyme and substrate respectively.
ES is an enzyme-substrate complex and P is the product.

Example – 30

Distinguish between true solution, colloidal solution and suspension.

Sol.

No.	True solution	Colloidal dispersion	Suspension
1.	It is homogenous.	It is heterogeneous.	It is heterogeneous.
2.	The particles are ions or molecules.	The particles are either single macromolecules or aggregates of molecules or atoms.	The particles are aggregates of molecules or atoms.
3.	The particle size ranges between 0.1 nm to 1nm.	The particle size lies between 1nm to 10^3 nm.	The particle size is larger than 10^3 nm.
4.	The particles do not settle.	The particles do not settle.	The particles do not settle out.
5.	The particles readily pass through parchment.	The particles pass slowly through parchment.	The particles do not pass through parchment.
6.	The particles are invisible even under powerful microscope	The particles are invisible under microscope.	The particles are visible under microscope or sometimes with naked eye.
7.	The particles do not scatter light	The particles scatter light.	It does not show the scatter light.
8.	It does not show electrophoresis, Brownian movement etc.	It shows the properties like Brownian movement, electrophoresis etc.	It does not show the properties electrophoresis, Brownian movement, etc.

Example – 31

Classify colloids on the basis of number of molecules or atoms.

Sol. On the basis of number of molecules or atoms colloidal systems are classified as Multimolecular and macromolecular colloids.

(i) Multimolecular Colloids :

(a) Colloidal particles are bigger than molecule of true solutions and smaller than the particles of suspensions.

(b) When a substance is dispersed in a suitable dispersion medium, a larger number of atoms or molecules aggregate to form a particle of colloidal size.

Example :

Au sol, Ag sol and Sulphur sol. Gold sol consists of particles which are formed by aggregation of many gold atoms.

(ii) Macromolecular Colloids

Certain substances form larger molecules having the dimension of colloidal particles. Such molecules are called macromolecules. Macromolecules have very high molecular masses.

Most lyophilic sols are macromolecular colloids.

Example :

gelatin, starch, nucleic acids.

Characteristics of macromolecular colloids :

(a) Some of the macromolecules dissolve in water and give homogenous solution.

(b) Solution of macromolecules behave like lyophilic colloids.

(c) There is considerable interaction between the solute and the solvent in macromolecular sols.

(d) Due to the large size and shape of the macromolecules, the macromolecular colloids cause serious deviations from the solutions obeying ordinary laws.

Example – 32

Write a note on optical properties of colloids (Tyndall effect).

Sol. It was investigated by British physicist John Tyndall.

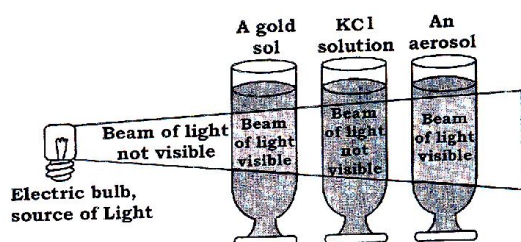
Colloidal system consist of discrete particles of dimensions comparable to the wavelength of the Ultraviolet and visible radiations So, whenever a beam of light is passed through a colloidal system, the colloidal particles scatter the light in all directions.

Now if the particle is viewed through a microscope, against a dark background, it becomes visible as a speck of light. This Scattering of light by the colloidal particles is known as Tyndall effect.

Tyndall effect is more prominent when the wavelength of the incident light and the dimensions of the

scattering particles are of the same order. That is why light is not scattered by true solutions. If the colloidal particles are assumed to be small, and do not absorb light, it may be observed that

- The intensity of the scattered light is greater in the direction parallel to the incident beam and smallest in the direction perpendicular to the incident beam.
- The intensity of the scattered light decreases and hence the tyndall beam assumes a blue colour imparting the complementary yellow colour to the transmitted light.

**Example – 33**

Write a note on electrical properties of colloids.

- Sol.**
- Electrical properties of colloidal dispersions are due to existence of electrical charge on colloidal particles.
 - All particles in a given dispersion carry same charge either +ve or -ve.
 - The repulsion between the particles carrying the same charge keeps them separated.

Due to the same charge the particles do not come close together to form large particles that will precipitate out. Thus stability of colloids is associated with an electrical charge present on the particles.

Example – 34

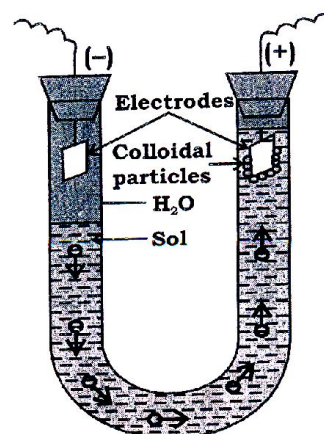
What are the various reasons for the origin of electrical charge on colloidal particles ?

- Sol.**
- The various reasons for the origin of electrical charge on the colloidal particles are as follows :
- Frictional electrification caused by the mutual rubbing of the colloidal particles with molecules of the dispersion medium.
 - Electron capture by particles from air and during electro-dispersion in Bredig's method.
 - Preferential adsorption of ions from solutions.

Example – 35

What happens in electrophoresis ?

- Sol.**
- The movement of colloidal particles under the influence of an electric field is called electrophoresis or cataphoresis.
 - The term cataphoresis is used because most of the colloidal sol studied at that time contained positively charged colloidal particles and thus migrated towards cathode.
 - The phenomenon of electrophoresis can be demonstrated by an apparatus shown below



Electrophoresis

Construction : The apparatus consists of a U-tube filled with colloidal sol. Water is added slowly over the sol to form sharp boundaries between sol and water in the two arms of the tube.

Two platinum electrodes are inserted in water in two arms of the U-tube and are connected to a high voltage battery.

Working : When an electric field is applied the boundary in one arm is seen to move down and in other arm to move up. If it is observed that the boundary falls gradually on the negative electrode side and moves up the positive electrode side, then the particles migrate towards positive electrode and they are negatively charged.

If the reverse occurs, then the particles carry positive charge.

Example – 36

Discuss Hardy – Schulze rule.

- Sol.** **Hardy – Schulze rules :**

The precipitation action of ions of an electrolyte depends on the sign of charge and the valence of the ion. The effects of these factors on the coagulation of

colloidal particles were stated by H. Schulze and Sir William Bate Hardy which are known as Schulze – Hardy rules.

(i) The ions of opposite sign to those present on the surface of the particles cause the precipitation of the particles. For example, positively charged ions of an electrolyte will cause the precipitation of negatively charged ions will effect the precipitation of positive particles.

(ii) The precipitation power of an electrolyte increases very rapidly with an increase in the valence of anion or cation. Thus, the precipitating power of ions decreases in the order $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$ and $\text{PO}_4^{3-} > \text{SO}_4^{2-} > \text{Cl}^-$.

Lyophilic colloids differ from lyophobic colloids in respect of coagulation by the addition of electrolytes. The lyophilic colloids require much larger amount for their precipitation than that required for the precipitation of lyophobic colloids. This is because the lyophilic particles are surrounded by layer of medium through which penetration of ions is difficult. The higher concentration of electrolytes remove this layer of medium surrounding the particles and then neutralises their charge resulting in their coagulation.

(v) An addition of solvents is a method useful for the precipitation of hydrophilic colloids. If solvents like alcohol or acetone, which have affinity for water are added to hydrophilic colloids, the colloids, the colloidal particles are coagulated due to dehydration.

Example – 37

Distinguish between oil in water and water in oil emulsions.

Sol.

No.	Oil in water		Water in oil
1.	Oil is the dispersed phase and water is the dispersion medium	1.	Water is the dispersed phase and oil is the dispersion medium
2.	If water is added, it will be miscible with the emulsion but if oil is added, it will not be miscible.	2.	If water is added, it will not be miscible with the emulsion but if oil is added, it will be miscible.
3.	An addition of small amount of an electrolyte makes the emulsion conducting.	3.	An addition of small amount of an electrolyte has not effect on conducting power.
4.	Water is continuous phase.	4.	Oil is continuous phase.
5.	Basic metal sulphates, water soluble alkali metal soaps are used as emulsifiers.	5.	Water insoluble soaps such as those of Zn, Al, Fe, alkaline earth metals are used as emulsifiers.

Example – 38

Comment on the statement that “Colloid is not a substance but state of a substance.”

Sol. The given statement is true. This is because the same substance may exist as a colloid under certain conditions and as a crystalloid under certain other conditons. For example, NaCl in water behaves as a crystalloid while in benzene, it behaves as a colloid. Similarly, dilute soap solution behaves like a crystalloid while concentrated solution behaves as a colloid (called associated colloid). It is the size of the particles which matters, i.e., the state in which the substance exists. If the size of the particles lies in the range 1 nm to 1000 nm, it is in the colloidal state.

Example – 39

1 g of charcoal adsorbs 100 mL 0.5 M CH_3COOH to form a monolayer, and thereby the molarity of CH_3COOH reduces to 0.49. Calculate the surface area of the charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = $3.0 \times 10^2 \text{ m}^2/\text{g}$.

Sol. Number of moles of acetic acid which we present in 100 mL before adding charcoal = 0.05

Number of moles of acetic acid in 100 mL after adding charcoal = 0.049

Number of moles of acetic acid adsorbed on the surface of charcoal = $(0.05 - 0.049) = 0.001$

So, number of molecules of acetic acid adsorbed on the surface of charcoal = $0.001 \times 6.02 \times 10^{23}$

$$= 6.02 \times 10^{20}$$

$$\therefore \text{Surface area of charcoal} = 3.01 \times 10^2 \text{ m}^2$$

Hence, area occupied by one molecule of acetic acid on the surface of charcoal

$$= \frac{3.01 \times 10^2}{6.02 \times 10^{20}} = 5 \times 10^{-19} \text{ m}^2,$$

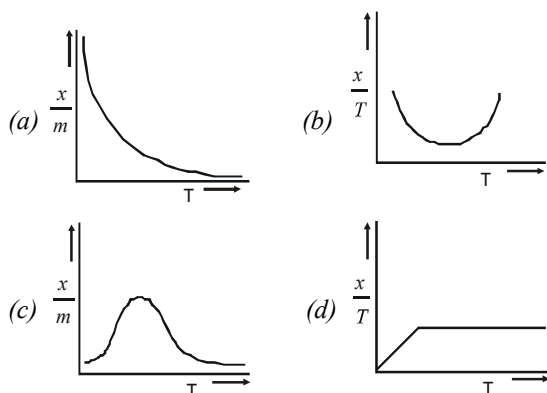
$$\text{So, area} = 5 \times 10^{-19} \text{ m}^2.$$

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Adsorption

- In adsorption of oxalic acid on activated charcoal, the activated charcoal is known as
(a) Adsorbent (b) Adsorbate
(c) Adsorber (d) Absorber
- Adsorption is phenomenon in which a substance
(a) Goes into the body of the other substance
(b) Remains close to the other substance
(c) Accumulates on the surface of the other substance
(d) None of these
- Adsorption is always
(a) Endothermic (b) Exothermic
(c) Either (a) or (b) (d) None of these
- Adsorption is
(a) Exothermic process hence increase in temperature decreases adsorption in cases where Vander Waal's forces exist between adsorbate and adsorbent.
(b) An endothermic process hence increase in temperature increases adsorption
(c) An exothermic process hence increase in temperature increases adsorption
(d) None of these
- There is desorption of physical adsorption when
(a) Temperature is increased
(b) Temperature is decreased
(c) Pressure is increased
(d) Concentration is increased
- Sorption is the term used when
(a) Adsorption takes place (b) Absorption takes place
(c) Both (a) and (b) (d) Desorption takes place
- Adsorption is multilayered in the case of
(a) Physical adsorption (b) Chemisorption
(c) Both (d) None of both
- Which of the following is not a characteristic of chemisorption
(a) ΔH is of the order of 400 kJ
(b) Adsorption is irreversible
(c) Adsorption may be multimolecular layer
(d) Adsorption is specific
- If the adsorbate is held on a surface by weak Vander Waal's forces, the adsorption process is called
(a) Physical adsorption
(b) Chemical adsorption
(c) Heat of adsorption
(d) Enthalpy of adsorption
- Physical adsorption is essentially quite appreciable
(a) At room temperature
(b) At higher temperature
(c) At lower temperature
(d) None of these
- Chemisorption
(a) Increases with temperature
(b) Decreases with temperature
(c) Remains unaffected by change of temperature
(d) first increases then decreases with temperature
- Physical adsorption
(a) Involves the weak attractive interaction between the adsorbent and adsorbate
(b) Involves the chemical interactions between the adsorbent and adsorbate
(c) Is irreversible in nature
(d) Increases with increase of temperature
- Chemisorption
(a) Involves the weak attractive interactions between adsorbent and adsorbate
(b) Is irreversible in nature
(c) Decreases with increase of temperature
(d) Involves multilayer formation of adsorbent on adsorbate
- When the temperature is lowered and pressure is raised, the adsorption of a gas on a solid
(a) Decreases
(b) Increases
(c) Remains unaffected
(d) Decreases first then increases

15. Which plot is the adsorption isobar for chemisorption where x is the amount of gas adsorbed on mass m (at constant pressure) at temperature T



16. For adsorption, the thermodynamic requirement is that

- (a) ΔH must be negative (b) ΔS must be negative
(c) ΔG must be negative
(d) ΔH , ΔS and ΔG must be negative

17. Which one of the following statement is not correct

- (a) The extent of adsorption depends on the nature of the adsorbent and adsorbate
(b) The extent of adsorption depends on the pressure of the gas
(c) The extent of adsorption depends on the temperature
(d) The extent of adsorption has no upper limit

18. Which one is Freundlich's equation

- (a) $\frac{x}{m} = \log k + \frac{1}{n} \log P$ (b) $\frac{x}{m} = \exp(-kP)$
(c) $\frac{x}{m} = kP^2$ (d) $\log \frac{x}{m} = \log k + \frac{1}{n} \log C$

Catalysis

19. Which of the following statements about a catalyst is true

- (a) It lowers the energy of activation
(b) The catalyst altered during the reaction is regenerated
(c) It does not alter the equilibrium
(d) All of these

20. A catalyst

- (a) Increases the free energy change in the reaction
(b) Decreases the free energy change in the reaction
(c) Does not increase or decrease the free energy change in the reaction
(d) Can either increase or decrease the free energy change depending on what catalyst we use

21. Which one is *false* in the following statements

- (a) A catalyst is specific in its action
(b) A very small amount of the catalyst alters the rate of a reaction
(c) The number of free valencies on the surface of the catalyst increases on subdivision
(d) Ni is used as catalyst in the manufacture of ammonia

22. A catalyst is a substance which

- (a) Alters the equilibrium in a reaction
(b) Is always in the same phase as the reactants
(c) Participates in the reaction and provides easier pathway for the same
(d) Does not participate in the reaction but speeds it up

23. Addition of catalyst in a system

- (a) Increases equilibrium concentrations
(b) No effect on equilibrium concentrations
(c) Decreases equilibrium concentrations
(d) Increases rate of forward reaction and decreases rate of backward reaction

24. In a homogeneous catalysis

- (a) The catalyst and the reactants should be gases
(b) The catalyst and the reactants should form a single phase
(c) Catalyst and the reactants are all solids
(d) The catalyst and the reactants are all liquids

25. Enzymes are

- (a) Micro-organisms (b) Proteins
(c) Inorganic compounds (d) Moulds

26. Enzymes are

- (a) Substances made by chemists to activate washing powder
(b) Very active vegetable catalysts
(c) Catalysts found in organisms
(d) Synthetic catalysts

27. In the human body, enzyme that catalyses the reaction of CO_2 with H_2O is
- (a) Invertase (b) Urease
(c) Nitrogenase (d) Carbonic anhydrate
28. 1. Following are the terms about activity and selectivity
- I: Activity is the ability of catalysts to accelerate chemical reactions and selectivity is the ability of catalysts to direct reaction to yield particular products
- II: Activity is the ability of catalysts to direct reaction to yield particular products and selectivity is the ability of catalysts to accelerate chemical reactions.
- Select correct term :
- (a) I (b) II
(c) I and II both (d) None of these
- Colloids**
29. Difference between colloids and crystalloids is of
- (a) Particle composition (b) Particle size
(c) Concentration (d) Ionic character
30. The shape of colloidal particles is
- (a) Sphere like (b) Rod like
(c) Disc like
(d) All the above
31. Which characteristic is true in respect of colloidal solutions
- (a) They always have two phases
(b) They are only in liquid state
(c) They can't be electrolysed
(d) They are only hydrophilic
32. Smoke is an example of
- (a) Gas dispersed in liquid (b) Gas dispersed in solid
(c) Solid dispersed in gas (d) Solid dispersed in solid
33. Milk is a colloid in which
- (a) A liquid is dispersed in liquid
(b) A solid is dispersed in liquid
(c) A gas is dispersed in liquid
(d) Some sugar is dispersed in water
34. When dispersed phase is liquid and dispersion medium is gas, then the colloidal system is called
- (a) Smoke (b) Clouds
(c) Emulsion (d) Jellies
35. When dispersion medium is water, the colloidal system is called
- (a) Sol (b) Aerosol
(c) Organosol (d) Aquasol
36. Butter is a colloid. It is formed when
- (a) Fat is dispersed in solid casein
(b) Fat globules are dispersed in water
(c) Water is dispersed in fat
(d) Casein is suspended in H_2O
37. Fog is an example of colloidal system
- (a) Liquid dispersed in gas (b) Gas dispersed in gas
(c) Solid dispersed in gas (d) Gas dispersed in liquid
38. A colloidal system in which gas bubbles are dispersed in a liquid is known as
- (a) Foam (b) Sol
(c) Aerosol (d) Emulsion
39. In shaving cream, the dispersion medium is
- (a) Liquid (b) Gas
(c) Solid (d) None of the above
40. Colloidal solution cannot be obtained from two such substances which are
- (a) Insoluble in each other
(b) In same physical state
(c) In different physical state
(d) None of the above
41. The difference between a lyophilic and lyophobic colloid is in their
- (a) Particle size
(b) Behaviour towards dispersion medium
(c) Filtrability (d) None of these
42. Which of the following statement is wrong for lyophobic sol
- (a) Dispersed phase is generally in organic material
(b) Can be easily coagulated by small addition of electrolyte
(c) Dispersed phase particles are poorly hydrated and colloid is stabilised due to charge on the colloidal particles
(d) Reversible in nature that is after coagulation can be easily set into colloidal form

43. When sugar is added to a colloidal solution it brings about
(a) Ionization (b) Coagulation
(c) Peptization (d) None of the above
44. Colloidal solutions of metals like gold, silver and platinum are generally prepared by using
(a) Peptization (b) Bredig's arc method
(c) Exchange of solvent (d) Oxidation method
45. When a freshly precipitated substance is converted into a colloidal solution with the help of a third substance, the process is known as
(a) Coagulation (b) Peptization
(c) Electrodispersion (d) Dialysis
46. Dialysis is the process of separation of
(a) Suspended particles from colloids
(b) Suspended particles from crystalloids
(c) Colloidal particles from crystalloids
(d) Colloidal particles from gel
47. Which one of the sols acts as protective colloid
(a) As_2S_3 (b) Gelatin
(c) Au (d) $Fe(OH)_3$
48. Milk is
(a) Dispersed fats in oil
(b) Dispersed fats in water
(c) Dispersed water in fats
(d) Dispersed water in oil
49. A cleared solution is again converted into colloidal solution. The process is called
(a) Peptisation (b) Electrolytic addition
(c) Electrophoresis (d) None of these
50. Following are various types of colloids. Match column *X* with column *Y*
- | <i>X</i> (Colloids) | | <i>Y</i> (Classification) | |
|---------------------|------------------|---------------------------|------------------|
| I | Rain cloud | - | <i>A</i> sol |
| II | Milk of magnesia | - | <i>B</i> aerosol |
| III | Soap suds | - | <i>C</i> gel |
| IV | Butter | - | <i>D</i> foam |
- Correct matching is
- | | | | |
|--------------|----------|----------|----------|
| I | II | III | IV |
| (a) <i>A</i> | <i>B</i> | <i>C</i> | <i>D</i> |
| (b) <i>A</i> | <i>C</i> | <i>B</i> | <i>D</i> |
| (c) <i>B</i> | <i>A</i> | <i>D</i> | <i>C</i> |
| (d) <i>B</i> | <i>A</i> | <i>C</i> | <i>D</i> |

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

- Which one of the following characteristics is not correct for physical adsorption ? (2003)
 - Adsorption on solids is reversible
 - Adsorption increases with increase in temperature
 - Adsorption is spontaneous
 - Both enthalpy and entropy of adsorption are negative
- The volume of a colloidal particle, V_c as compared to the volume of a solute particle in a true solution V_s , could be (2005)
 - $\frac{V_c}{V_s} \approx 10^3$
 - $\frac{V_c}{V_s} \approx 10^{-3}$
 - $\frac{V_c}{V_s} \approx 10^{23}$
 - $\frac{V_c}{V_s} \approx 1$
- The disperse phase in colloidal iron (III) hydroxide and colloidal gold is positively and negatively charged, respectively. Which of the following statement is not correct ? (2005)
 - Coagulation in both sols can be brought about by electrophoresis
 - Mixing the sols has no effect
 - Sodium sulphate solution causes coagulation in both sols
 - Magnesium chloride solution coagulates, the gold sol more readily than the iron (III) hydroxide sol
- In Langmuir's model of adsorption of a gas on a solid surface (2006)
 - the rate of dissociation of adsorbed molecules from the surface does not depend on the surface covered
 - the adsorption at a single site on the surface may involve multiple molecules at the same time
 - the mass of gas striking a given area of surface is proportional to the pressure of the gas
 - the mass of gas striking a given area of surface is independent of the pressure of the gas
- Gold numbers of protective colloids A, B, C and D are 0.50, 0.01, 0.10 and 0.005, respectively. The correct order of their protective powers is (2008)
 - $D < A < C < B$
 - $C < B < D < A$
 - $A < C < B < D$
 - $B < D < A < C$
- Which of the following statements is incorrect regarding physisorptions ? (2009)
 - It occurs because of van der Waals' forces
 - More easily liquefiable gases are adsorbed readily
 - Under high pressure it results into multi molecular layer on adsorbent surface
 - Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is slow and positive
- According to Freundlich adsorption isotherm which of the following is correct ? (2012)
 - $\frac{x}{m} \propto p^0$
 - $\frac{x}{m} \propto p^1$
 - $\frac{x}{m} \propto p^{1/n}$
 - All of the above are correct for different range of pressure
- The coagulating power of electrolytes having ions Na^+ , Al^{3+} and Ba^{2+} for arsenic sulphide sol increases in the order (2013)
 - $\text{Al}^{3+} < \text{Ba}^{2+} < \text{Na}^+$
 - $\text{Na}^+ < \text{Ba}^{2+} < \text{Al}^{3+}$
 - $\text{Ba}^{2+} < \text{Na}^{2+} < \text{Al}^{3+}$
 - $\text{Al}^{3+} < \text{Na}^+ < \text{Ba}^{2+}$
- 3 g of activated charcoal was added to 50 mL of acetic acid solution (0.06N) in a flask. After an hour it was filtered and the strength of the filtrate was found to be 0.042 N. The amount of acetic acid adsorbed (per gram of charcoal) is : (2015)
 - 42 mg
 - 54 mg
 - 18 mg
 - 36 mg
- For a linear plot of $\log (x/m)$ versus $\log p$ in a Freundlich adsorption isotherm, which of the following statement is correct? (k and n are constants) (2016)
 - $1/n$ appears as the intercept
 - Only $1/n$ appears as the slope.
 - $\log (1/n)$ appears as the intercept
 - both k and $1/n$ appear in the slope term.
- Adsorption of a gas on a surface follows Freundlich adsorption isotherm. Plot of $\log \frac{x}{m}$ versus $\log p$ gives a straight line with slope equal to 0.5, then : (2017)

($\frac{x}{m}$ is the mass of the gas adsorbed per gram of adsorbent)

 - Adsorption is independent of pressure.
 - Adsorption is proportional to the pressure.
 - Adsorption is proportional to the square root of pressure.
 - Adsorption is proportional to the square of pressure.

JEE MAINS ONLINE QUESTION

1. Which one of the following is an incorrect statement for physisorption **Online 2015 SET (1)**

- (a) It is a reversible process
- (b) It requires less heat of adsorption
- (c) It requires activation energy
- (d) It takes place at low temperature

2. A particular adsorption process has the following characteristics : (i) It arises due to van der Waals forces and (ii) it is reversible. Identify the correct statement that describes the above adsorption process :

Online 2016 SET (1)

- (a) Enthalpy of adsorption is greater than 100 kJ mol^{-1} .
- (b) Energy of activation is low.
- (c) Adsorption is monolayer.
- (d) Adsorption increases with increase in temperature.

3. Gold numbers of some colloids are : Gelatin : 0.005 – 0.01, Gum Arabic : 0.15 – 0.25; Oleate : 0.04 – 1.0; Starch: 15 – 25. Which among these is a better protective colloid ?

Online 2016 SET (2)

- (a) Gelatin (b) Gum Arabic
- (c) Oleate (d) Starch

4. Among the following, **correct** statement is :

Online 2017 SET (1)

- (a) Brownian movement is less pronounced for smaller particles than for bigger-particles.
- (b) Sols of metal sulphides are lyophilic.
- (c) Hardy Schulze law states that bigger the size of the ions, the greater is its coagulating power.
- (d) One would expect charcoal to adsorb SO_2 more than hydrogen CH_4 .

5. Adsorption of a gas on a surface follows Freundlich adsorption isotherm. Plot of $\log \frac{x}{m}$ versus $\log p$ gives a straight line with slope equal to 0.5, then :

($\frac{x}{m}$ is the mass of the gas adsorbed per gram of adsorbent) **Online 2017 SET (2)**

- (a) Adsorption is independent of pressure.
- (b) Adsorption is proportional to the pressure.
- (c) Adsorption is proportional to the square root of pressure.
- (d) Adsorption is proportional to the square of pressure.

6. Which of the following statements about colloids is False ?

Online 2018 SET (1)

- (a) Freezing point of colloidal solution is lower than true solution at same concentration of a solute.
- (b) Colloidal particles can pass through ordinary filter paper.
- (c) When silver nitrate solution is added to potassium iodide solution, a negatively charged colloidal solution is formed.
- (d) When excess of electrolyte is added to colloidal solution, colloidal particle will be precipitated.

7. If x gram of gas is adsorbed by m gram of adsorbent at

pressure P, the plot of $\log \frac{x}{m}$ versus $\log P$ is linear. The

slope of the plot is : **Online 2018 SET (2)**

(n and k are constants and $n > 1$)

- (a) 2k (b) $\log k$

- (c) n (d) $\frac{1}{n}$

8. Which one of the following is not a property of physical adsorption ? **Online 2018 SET (3)**

- (a) Higher the pressure, more the adsorption
- (b) Lower the temperature, more the adsorption
- (c) Greater the surface area, more the adsorption
- (d) Unilayer adsorption occurs

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Adsorption

1. (S) The adsorption of a gas on a solid surface varies with pressure of the gas in which of the following manner
- Fast $\rightarrow$ slow independent of the pressure
 - Slow fast independent of the pressure
 - Independent of the pressure fast slow
 - Independent of the pressure slow fast

2. (S) For the adsorption of a gas on a solid, the plot of $\log(x/m)$ versus $\log P$ is linear with slope equal to
- k
 - $\log k$
 - n
 - $1/n$

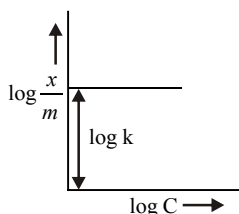
3. (S) In Freundlich adsorption isotherm, adsorption is proportional to pressure P as
- P^0
 - P
 - P^n
 - $P^{1/n}$

4. (S) For the adsorption of solution on a solid surface
- $$\frac{x}{m} = kC^{1/n}$$

Adsorption isotherm of $\log\left(\frac{x}{m}\right)$ and $\log C$ was found of the type

This is when

- $C = 0$
- $C = 1 M$
- $C = \text{constant}$
- $C = 2 M$



5. (S) Following are the events taking place to explain adsorption theory

I : Desorption

II : Diffusion of the reactants along the surface

III : Adsorption of the reactants

IV : Formation of the activated surface complex

These events are taking place in the following order

- I, II, III, IV
- III, II, IV, I
- III, IV, I, II
- IV, III, II, I

6. (S) There are certain properties related to physical adsorption

I : Reversible

II : Formation of unimolecular layer

III : Low heat of adsorption

IV : Occurs at low temperature and decreases with increasing temperature

Which of the above properties are for physical adsorption

- I, II, III
- I, III, IV
- II, III, IV
- I, III

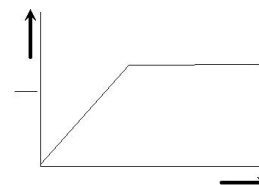
7. (S) In the following isotherm

- $\frac{x}{m} \propto P^0$ when point A is reached

- Desorption may start along AB

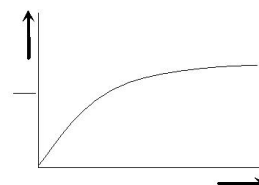
- $\frac{x}{m} \propto P^{1/n}$ along OA

- All of these

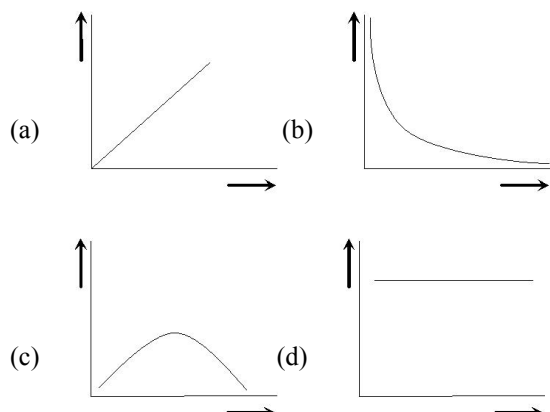


8. (S) When saturation is attained in terms of adsorption, variation of $\left(\frac{x}{m}\right)$ and C (concentration) is given by the portion of the isotherm

- OA
- OB
- AB
- BC



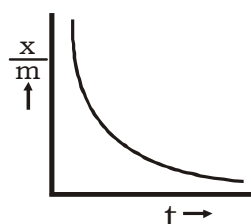
9. (S) Following is the variation of physical adsorption with temperature



10. (S) Rate of physisorption increases with

- (a) decrease in temperature
(b) increases in temperature
(c) decrease in pressure
(d) decrease in surface area

Rate of physisorption increases with decrease of temperature.



where x/m = mass of gas adsorbed per unit mass of adsorbent and t = temperature.

11. (S) Adsorption of gases on solid surface is generally exothermic because

- (a) enthalpy is positive (b) entropy decreases
(c) entropy increases (d) free energy increases

When the gas is adsorbed on surface, the freedom of movement of gaseous molecules becomes restricted. (This causes decrease in the entropy of the gas after adsorption, i.e., Δ becomes negative.) Hence, ΔS becomes negative.

12. (S) Graph between $\log \frac{x}{m}$ and $\log P$ is a straight line inclined at an angle $\theta = 45^\circ$. When pressure is 0.5 atm and $\log k = 0.699$, the amount of solute adsorbed per g of adsorbent will be
- (a) 1 g/g adsorbent (b) 1.5 g/g adsorbent
(c) 2.5 g/g adsorbent (d) 0.25 g/g adsorbent

13. (M) Select the correct statements among following:
- (a) At 83 K, N_2 is physisorbed on the surface of iron
(b) At 773 K and above N_2 is chemisorbed on the iron surface
(c) Activation energy is +ve in case of physisorption and zero in case of chemisorption
(d) Activation energy is zero in case of physisorption and +ve in case of chemisorption

14. (M) Which of the following is/are correct about Freundlich's adsorption isotherm for gases and solutions?

(a) $\frac{x}{m} = KC^{1/n}$ (b) $\frac{x}{m} = \frac{1+bP}{a}$

(c) $\log \left(\frac{x}{m} \right) = \log K + n \log P$

(d) $\log \left(\frac{x}{m} \right) = \log K + \frac{1}{n} \log C$

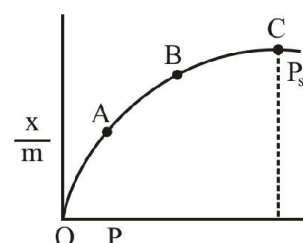
Comprehension

A graph between x/m and the pressure P of the gas at a constant temperature is called adsorption isotherm. Where x is the no. of moles of the adsorbate and m is the mass of the adsorbent. Adsorption isotherms of different shapes have been experimentally observed. According to Freundlich adsorption isotherm.

$$\frac{x}{m} = kP^{1/n}$$

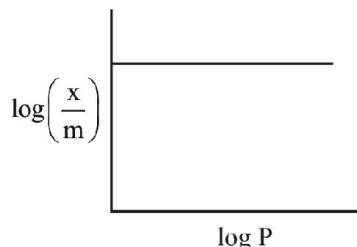
where k and n are constant parameters depending upon the nature of the solid and gas.

15. (C) In the given isotherm select the incorrect statement:



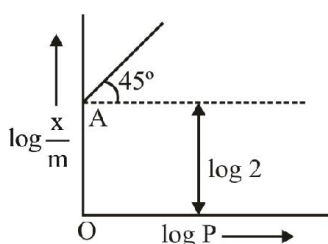
- (a) $\frac{x}{m} \propto P^{1/n}$ along OA
(b) $\frac{x}{m} \propto P^0$ when point B is reached
(c) $\frac{x}{m}$ does not increase as rapidly with pressure along BC due to less surface area available for adsorption
(d) nature of isotherm is different for two gases for same adsorbent

16. (C) Adsorption isotherm of $\log\left(\frac{x}{m}\right)$ and $\log P$ was found of the type:



This is true when:

- (a) $P = 0$ (b) $P = 1$
(c) $\frac{1}{n} = 1$ (d) $\frac{1}{n} = \infty$
17. (C) Graph between $\log\left(\frac{x}{m}\right)$ and $\log P$ is a straight line at angle 45° with intercept OA as shown.



Hence, $\left(\frac{x}{m}\right)$ at a pressure at 2 atm is:

- (a) 2 (b) 4
(c) 8 (d) 1
18. (A) **Assertion :** For adsorption ΔG , ΔH , ΔS all have -ve values.
Reason : Adsorption is a exothermic process in which randomness decreases due to force of attraction between adsorbent and adsorbate.
(a) A (b) B
(c) C (d) D
19. (A) **Assertion :** The extent of adsorption of CO_2 is much more higher than of H_2 .
Reason : $\text{CO}_2(\text{g})$ has higher critical temperature and more van der Waals' force of attraction as compare to $\text{H}_2(\text{g})$.
(a) A (b) B
(c) C (d) D

20. (X) Column-I

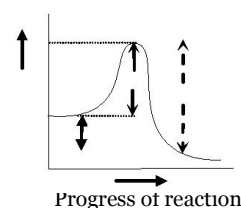
- (A) Chemisorption
(B) Physical adsorption
(C) Desorption

Column-II

- (p) Exothermic
(q) Endothermic
(r) Removal of adsorbed material
(s) Highly specific in nature

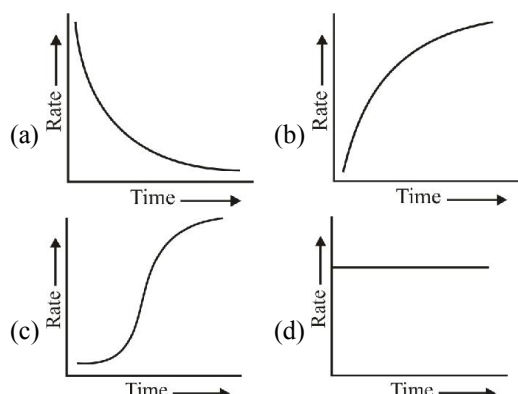
Catalysis

21. (S) Enzymes are known to increase the rate of reaction by
(a) 10^2 times (b) 10^{-2} times
(c) 10^5 times (d) 10^{12} times
22. (S) The ability of the catalyst to direct the reaction to yield particular product is called
(a) Reactivity (b) Selectivity
(c) Activity (d) Fugacity
23. (S) Enzymes are
(a) Micro-organisms (b) Protein
(c) Inorganic compounds
(d) Moulds
24. (S) Which of the following statements is incorrect
(a) Enzymes are in colloidal state
(b) Enzymes are catalysts
(c) Enzymes can catalyse any reaction
(d) Urease is an enzyme
25. (S) Energy of activation of forward and backward reaction are equal in cases (numerical values) where
(a) $\Delta H = 0$ (b) No catalyst present
(c) $\Delta S = 0$
(d) Stoichiometry is the mechanism
26. (S) Mark the correct statement about given graph



- (a) X is threshold energy level
(b) Y and Z are energy of activation for forward and backward reaction respectively
(c) Q is heat of reaction and reaction is exothermic
(d) All of these

27. (S) Oxidation of oxalic acid by acidified KMnO_4 is an example of autocatalysis. Which of the following graph represents the variation of rate of this reaction with time?



28. (S) Which of the following is/are not affected by the catalyst?

- (a) Heat of reaction (b) Equilibrium constant
(c) Amount of product (d) Rate constant of the reaction

29. (M) Which of the following are correct about the catalyst?

- (a) They participate in the reaction but recovered at last
(b) It does not affect ΔG
(c) It does not affect ΔH
(d) It alters the mechanism of reaction

30. (M) Which of the following act as negative catalyst?

- (a) Ethanol in oxidation of chloroform
(b) Tetra ethyl lead used as antiknocking agent
(c) Glycerol in the decomposition
(d) Fe in the formation of ammonia by Haber process

31. (A) **Assertion :** A reaction cannot become fast by itself unless a catalyst is added in the reaction mixture.

Reason : Catalyst may increase or decrease the rate of a reaction.

- (a) A (b) B
(c) C (d) D

32. (A) **Assertion :** Positive catalysts increase the rate of chemical reaction.

Reason : Catalysts increase the rate of reaction by lowering the activation energy of the reaction.

- (a) A (b) B
(c) C (d) D

33. (X) **Column-I**

(A) BaSO_4

(B) Acetamide

(C) Zeolite

(D) Nickel

- Column-II**

(p) Inhibitor for decomposition of H_2O_2

(q) Catalyst

(r) Remove hardness of water

(s) Poison for Pd in Lindlar's catalyst

Colloids

34. (S) The simplest way to check whether a system is colloidal, is

- (a) Tyndall effect (b) Electro dialysis
(c) Brownian movement (d) Finding out particle size

35. (S) Blood may be purified by

- (a) Dialysis (b) Electro - osmosis
(c) Coagulation (d) Filtration

36. (S) Colloidal particles carrying charge is best shown by

- (a) Brownian movement
(b) Tyndall effect
(c) Electrophoresis (d) Dialysis

37. (S) As_2S_3 sol has a negative charge. Capacity to precipitate it is highest in

- (a) AlCl_3 (b) Na_3PO_4
(c) CaCl_2 (d) K_2SO_4

38. (S) Colloidal solution is not purified by

- (a) Dialysis (b) Electrodialysis
(c) Electrophoresis (d) Ultrafiltration

39. (S) Peptising agent is

- (a) Always an electrolyte
(b) Always a non - electrolyte
(c) Electrolyte or non - electrolyte
(d) A lyophilic colloid

40. (S) Coagulation or demulsification can be done by some of the methods given below

I : By addition of substance which would destroy the emulsifier

II : By addition of an electrolyte which would destroy the charge

III : By heating, freezing and centrifuging

Select correct methods

- (a) I, II (b) I, II, III
(c) II only (d) III only

41. (S) Sky looks blue due to

- (a) Dispersion effect (b) Reflection
(c) Transmission (d) Scattering

42. (S) Ferric hydroxide sol is positively charged colloid. The coagulating power of NO_3^- , SO_4^{2-} and PO_4^{3-} ions would be in the order

- (a) $\text{NO}_3^- > \text{SO}_4^{2-} > \text{PO}_4^{3-}$ (b) $\text{SO}_4^{2-} > \text{NO}_3^- > \text{PO}_4^{3-}$
(c) $\text{PO}_4^{3-} > \text{SO}_4^{2-} > \text{NO}_3^-$ (d) $\text{NO}_3^- = \text{SO}_4^{2-} = \text{PO}_4^{3-}$

43. (S) Maximum coagulation power is in

- (a) Na^+ (b) BA^{++}
(c) Al^{+++} (d) Sn^{++++}

44. (S) The colloidal particles can pass through
(a) Filter paper as well as animal membrane
(b) Animal membrane but not through filter paper
(c) Filter paper but not through animal membrane
(d) Semipermeable membrane
45. (S) Which of the following is most effective in coagulating a ferric hydroxide sol
(a) KCl (b) KNO_3
(c) K_2SO_4 (d) $K_3[Fe(CN)_6]$
46. (S) In neutralisation of KI by $AgNO_3$ positive charge is due to absorption of
(a) Ag^+ ions (b) Ag
(c) I^- ions (d) Both (b) and (c)
47. (S) Cleansing action of soap occurs because
(a) Oil and grease can be absorbed into the hydrophobic centres of soap micelles and washed away
(b) Oil and grease can be absorbed into the hydrophilic centres of soap micelles and washed away
(c) Oil and grease can be absorbed into both hydrophilic and hydrophobic centres but not washed away
(d) Cleansing action is not related to micelles
48. (S) Colloidal paints are used to paint metallic portions of cars. This application is based on
(a) Coagulation (b) Electro-osmosis
(c) Peptization (d) Electrophoresis
49. (S) Ultramicroscope is used for observing
(a) Light reflection (b) Light absorption
(c) Light scattering (d) Light polarisation
50. (S) Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient conditions is
(a) $CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$
(b) $CH_3(CH_2)_{11}OSO_3^-Na^+$
(c) $CH_3(CH_2)_6COO^-Na^+$
(d) $CH_3(CH_2)_{11}N^+(CH_3)_3Br^-$
Option (a) and (b) both are capable of forming micelle
(a) $\rightarrow CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$
Cetyl trimethyl ammonium bromide
(b) $\rightarrow CH_3(CH_2)_{11}OSO_3^-Na^+$
Sodium lauryl sulphate
But critical concentration for micelle formation decreases as the molecular weight of hydrocarbon chain of surfactant grows because in this true solubility diminishes and the tendency of surfactant molecule to undergo association increases.
51. (M) Select the correct statements for Brownian movement:
(a) It becomes more intense with increase in temperature
(b) It increases with increase in particle size
(c) It is due to collision of molecules of dispersion medium with the colloidal particles
(d) It increases the stability of colloidal solution
52. (M) Select the correct statements among following:
(a) Foam is a colloidal solution of gas in liquid
(b) Aerosol is a colloidal solution of liquid in gas
(c) Gel is a colloidal solution of solid in liquid
(d) Emulsion is a colloidal solution of liquid in liquid
53. (M) Which of the following statements are true?
(a) Flocculation value is inversely proportional to coagulating power
(b) Colloidal silica is protective colloid
(c) Alum is used for cleaning muddy water
(d) Gelatin is added in ice cream, it acts as emulsifier
54. (M) Select the correct statements about enzymes:
(a) Enzymes are biological catalysts found in organisms
(b) All enzymes are proteins
(c) Enzymes can catalyse any reaction
(d) Enzymes activity is optimum at $27^\circ C$
- ### Comprehension
- Coagulation is the process by which the dispersed phase of a colloid is made to aggregate and thereby separate from the continuous phase. The minimum concentration of an electrolyte in milli-moles per litre of the electrolyte solution which is required to cause the coagulation of colloidal sol is called coagulation value.
- There higher is the coagulating power of effective ion, smaller will be the coagulation value of the electrolyte.
- $$\text{Coagulation value} \propto \frac{1}{\text{coagulating power}}$$
- The coagulation values of different electrolytes are different. This behaviour can be easily understood by Hardy-Schulze rule which states.
- “The greater is the valency of the effective ion greater is its precipitating power.”
55. (C) Which one has the highest coagulation power?
(a) K^+ (b) Ca^{2+}
(c) Al^{3+} (d) Sn^{4+}
56. (C) As_2S_3 sol is negatively charged, capacity to precipitate it is highest in:
(a) K_2SO_4 (b) Na_3PO_4
(c) $AlCl_3$ (d) $CaCl_2$
57. (C) The ability of an ion to bring coagulation of a given colloid depends upon:
(a) the sign of its charge
(b) magnitude of its charge
(c) both magnitude and sign
(d) none of these

58. (C) The coagulation of colloidal particles of the sol can be caused by:
- (a) heating (b) adding electrolyte
- (c) adding oppositely charged sol
- (d) all of these

Comprehension

Emulsions are normally prepared by shaking the two components together vigorously although some kind of emulsifying agent usually has to be added to stabilize the product. This emulsifying agent may be a soap or other surfactant (surface active) species or a lyophilic sol that forms a protective film around the dispersed phase.

Emulsions broadly classified into two types:

- (i) Oil in water emulsions (O/W): Oil acts as dispersed phase and water acts as dispersion medium.
- (ii) Water in oil emulsions (W/O): Water acts as dispersed phase and oil acts as dispersion medium. Dye test, dilution test may be employed for identification of emulsions.

59. (C) Read two statements:

- (1) Milk is an example of oil in water (O/W) type emulsion
- (2) Cold cream is an example of water in oil (W/O) type emulsions
- (a) Only statement 1 is correct
- (b) Only statement 2 is correct
- (c) Both are correct
- (d) None of these

60. (C) Select correct statement:

- (a) Water in oil emulsions are more viscous than the aqueous emulsions
- (b) Electrical conductance of aqueous emulsions is less than that of oil emulsions
- (c) Deemulsification can be done by soap or detergent
- (d) An emulsion can be diluted with H_2O then it is oil in water (O/W) type

61. (A) **Assertion :** Micelles are formed by surfactant molecules above the critical micelle concentration (CMC).

Reason : The conductivity of a solution having surfactant molecules decrease sharply at the CMC.

- (a) A (b) B
- (c) C (d) D

62. (A) **Assertion :** For coagulation of positively charged sols, $[Fe(CN)_6]^{4-}$ ion has higher coagulating power than that of PO_4^{3-} , SO_4^{2-} , Cl^- .

Reason : Because according to Hardy Schulze rule, higher is the valency of ions for the oppositely charged sol particles, better will be the precipitation.

- (a) A (b) B
- (c) C (d) D

63. (X) **Column-I** **Column-II**
- (A) Liquid dispersed in gas (p) Foam
- (B) Gas dispersed in liquid (q) Emulsion
- (C) Liquid dispersed in solid (r) Aerosol
- (D) Liquid dispersed in liquid (s) Gel

64. (X) **Column-I** **Column-II**
- (A) Milk (p) Aerosol
- (B) Dust (q) Emulsion
- (C) Cheese (r) Gel
- (D) Froth (s) Foam

65. (X) **Column-I** **Column-II**
- (A) Peptization (p) Preparation of sols
- (B) Ultra centrifugation (q) Purification of sols
- (C) Electrodialysis (r) Preparation of metal sols
- (D) Bredig's method (s) Movement of ion across the membrane in presence of electric field

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

- Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at 25°C. For this process, the correct statement is (2013)
 - the adsorption requires activation at 25°C
 - the adsorption is accompanied by a decrease in enthalpy
 - the adsorption increases with increase of temperature
 - the adsorption is irreversible
- Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}_2(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulating agent for Sb_2S_3 is (2009, 1M)
 - Na_2SO_4
 - CaCl_2
 - $\text{Al}_2(\text{SO}_4)_3$
 - NH_4Cl
- Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient conditions, is (2008, 3M)
 - $\text{CH}_3(\text{CH}_2)_{15}\text{N}^+(\text{CH}_3)_3\text{Br}^-$
 - $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^-\text{Na}^+$
 - $\text{CH}_3(\text{CH}_2)_6\text{COO}^-\text{Na}^+$
 - $\text{CH}_3(\text{CH}_2)_{11}\text{N}^+(\text{CH}_3)_3\text{Br}^-$
- Spontaneous adsorption of a gas on solid surface is an exothermic process because (2007, 3M)
 - ΔH increases for system
 - ΔS increases for gas
 - ΔS decreases for gas
 - ΔG increases for gas
- Lyophilic sols are (2005, 1M)
 - irreversible sols
 - prepared from inorganic compounds
 - coagulated by adding electrolytes
 - self-stabilising
- Rate of physisorption increases with (2003, 1M)
 - decrease in temperature
 - increase in temperature
 - decrease in pressure
 - decrease in surface area
- Choose the correct reason(s) for the stability of the lyophobic colloidal particles. (2012)
 - Preferential adsorption of ions on their surface from the solution
 - Preferential adsorption of solvent on their surface from the solution.
 - Attraction between different particles having opposite charges on their surface.
 - Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles.
- The given graph/data I, II, III and IV represent general trends observed for different physisorption and chemisorption processes under mild conditions of temperature and pressure. Which of the following choice(s) about I, II, III and IV is (are) correct? (2012)

(I)

(II)

(III)

(IV)

 - I is physisorption and II is chemisorption
 - I is physisorption and III is chemisorption
 - IV is chemisorption and II is chemisorption
 - IV is chemisorption and III is chemisorption

9. The correct statement(s) pertaining to the adsorption of a gas on a solid surface is(are) (2011)
- Adsorption is always exothermic.
 - Physisorption may transform into chemisorption at high temperature
 - Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
 - Chemisorption is more exothermic than physisorption however it is very slow due to higher energy of activation.
10. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2 . The true statement(s) regarding this adsorption is (are) (2015)
- O_2 is physisorbed
 - Heat is released
 - Occupancy of π_{2p}^* of O_2 is increased
 - Bond length of O_2 is increased
11. The correct statement(s) about surface properties is(are) (2017)
- Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system
 - The critical temperatures of ethane and nitrogen are 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature
 - Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium
 - Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution

Assertion and Reason

Read the following questions and answer as per the direction given below:

- Assertion is true; Reason is true; Reason is a correct explanation of Assertion.
 - Assertion is true; Reason is true; Reason is not the correct explanation of Assertion.
 - Assertion is true; Reason is false.
 - Assertion is false; Reason is correct.
12. **Assertion :** Micelles are formed by surfactant molecules above the critical micelle concentration(CMC).
Reason : The conductivity of a solution having surfactant molecules decreases sharply at the CMC. (2007)
- A
 - B
 - C
 - D

ANSWER KEY

Exercise - 1 : (Basic Objective Questions)

1. (a)	2. (c)	3. (b)	4. (a)	5. (a)	6. (c)	7. (a)	8. (c)	9. (a)	10. (c)
11. (d)	12. (a)	13. (b)	14. (b)	15. (c)	16. (c)	17. (d)	18. (d)	19. (d)	20. (c)
21. (d)	22. (d)	23. (b)	24. (b)	25. (b)	26. (c)	27. (d)	28. (a)	29. (b)	30. (d)
31. (a)	32. (c)	33. (a)	34. (b)	35. (d)	36. (c)	37. (a)	38. (a)	39. (a)	40. (d)
41. (b)	42. (d)	43. (a)	44. (b)	45. (b)	46. (c)	47. (b)	48. (b)	49. (d)	50. (c)

Exercise - 2 : (Previous Year JEE Mains Questions)

1. (b)	2. (a)	3. (b)	4. (a)	5. (c)	6. (d)	7. (d)	8. (b)	9. (c)	10. (b)
11. (c)									

JEE Mains Online

1. (c)	2. (b)	3. (a)	4. (d)	5. (c)	6. (a)	7. (d)	8. (d)
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Exercise - 3 : (Advanced Objective Questions)

1. (a)	2. (d)	3. (d)	4. (b)	5. (b)	6. (b)	7. (d)	8. (d)	9. (b)	10. (a)
11. (b)	12. (c)	13. (abd)	14. (abd)	15. (b)	16. (c)	17. (b)	18. (a)	19. (a)	
20. (A – p, s; B – p; C – q, r; D – q)	21. (d)	22. (b)	23. (b)	24. (c)	25. (a)	26. (d)	27. (c)		
28. (abc)	29. (abcd)	30. (abc)	31. (d)	32. (a)	33. (A – s; B – p; C – q, r; D – q)	34. (a)	35. (a)		
36. (c)	37. (a)	38. (c)	39. (a)	40. (b)	41. (d)	42. (c)	43. (d)	44. (c)	45. (d)
46. (a)	47. (a)	48. (d)	49. (c)	50. (a)	51. (acd)	52. (abd)	53. (acd)	54. (abd)	55. (d)
56. (c)	57. (c)	58. (d)	59. (c)	60. (d)	61. (a)	62. (a)	63. (A – r; B – p; C – s; D – q)		
64. (A – q; B – p; C – r; D – s)	65. (A – p; B – q; C – q, s; D – p, r)								

Exercise - 4 : (Previous Year JEE Advanced Questions)

1. (b)	2. (c)	3. (a)	4. (c)	5. (d)	6. (a)	7. (ad)	8. (ac)	9. (a, b, d)	10. (b, c, d)
11. (a, b)	12. (b)								

Dream on !!

IMPORTANT INFORMATION

S.No.	Common Name	Chemical Name	Formulae
1.	Alum	Hydrated double sulphate of potassium and aluminium	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
2.	Alumina	Aluminium oxide	Al_2O_3
3.	Ammonia water	Ammonium hydroxide	NH_4OH
4.	Anglesite	Lead sulphate	$PbSO_4$
5.	Aqua fortis	Concentrated nitric acid	HNO_3
6.	Aqua regia		$3HCl(Conc) + HNO_3(Conc)$
7.	Azote gas	Nitrogen	N_2
8.	Azurite blue	Basic copper carbonate	$2CuCO_3 \cdot Cu(OH)_2$
9.	Baking Soda	Sodium hydrogen carbonate	$NaHCO_3$
10.	Baryta	Barium Hydroxide	$Ba(OH)_2$
11.	Basic lead acetate		$Pb(OH)_2 \cdot Pb(CH_3COO)_2$
12.	Bauxite	Dihydrate Aluminium oxide	$Al_2O_3 \cdot 2H_2O$
13.	Blue Vitriol	Pentahydrate of cupric sulphate	$CuSO_4 \cdot 5H_2O$
14.	Bleaching powder	Calcium oxychloride or calcium (hypochlorite) chloride	$CaOCl_2$ or $Ca(OCl)Cl$
15.	Borax	Sodium tetraborate	$Na_2B_4O_7 \cdot 10H_2O$
16.	Brimstone	Sulphur	S
17.	Brine or common salt or rock salt	Sodium Chloride	NaCl
18.	Cerussite	Lead carbonate	$PbCO_3$
19.	Chalk or limestone	Calcium Carbonate	$CaCO_3$
20.	Chile saltpetre or Caliche	Sodium nitrate	$NaNO_3$
21.	Cuprite or ruby copper	Cuprous oxide	Cu_2O
22.	Copper glance	Copper sulphide	Cu_2S
23.	Carnalite	Potassium magnesium chloride	$KCl \cdot MgCl_2 \cdot 6H_2O$
24.	Calomel	Mercurous chloride	Hg_2Cl_2
25.	Cane sugar or Beet sugar	Sucrose	$C_{12}H_{22}O_{11}$
26.	Calgon	Sodium hexa-metaphosphate	$Na_2[Na_4(PO_3)_6]$
27.	Caustic soda	Sodium hydroxide	NaOH
28.	Caustic potash	Potassium hydroxide	KOH
29.	Calamine	Zinc carbonate	$ZnCO_3$
30.	Corundum	Aluminium oxide	Al_2O_3
31.	Dead burnt gypsum	Anhydrous calcium sulphate	$CaSO_4$
32.	Diaspore	Monohydrate aluminium oxide	$Al_2O_3 \cdot H_2O$
33.	Dry ice	Solid carbon dioxide	CO_2 (solid)
34.	Epsom salt	Heptahydrate of magnesium sulphate	$MgSO_4 \cdot 7H_2O$
35.	Fluorspar	Calcium fluoride	CaF_2
36.	Galena	Lead sulphide	PbS
37.	Glauber's salt	Hydrated sodium sulphate	$Na_2SO_4 \cdot 10H_2O$
38.	Green Vitriol	Ferrous sulphate	$FeSO_4 \cdot 7H_2O$
39.	Gypsum	Calcium sulphate dihydrate	$CaSO_4 \cdot 2H_2O$
40.	Hydrolith	Calcium hydride	CaH_2
41.	Hypo	Sodium thiosulphate	$Na_2S_2O_3 \cdot 5H_2O$
42.	Iron pyrites	Iron disulphide	FeS_2
43.	Laughing gas	Dinitrogen oxide	N_2O
44.	Limonite	Hydrated ferric oxide	$Fe_2O_3 \cdot 3H_2O$
45.	Litharge	Lead monoxide	PbO
46.	Lime	Calcium oxide	CaO

47.	Lime water or slaked lime	Calcium hydroxide	Ca(OH)_2
48.	Lime stone or chalk or marble	Calcium carbonate	CaCO_3
49.	Lunar caustic	Silver nitrate	AgNO_3
50.	Magnetite	Ferrosoferric oxide	Fe_3O_4
51.	Milk of lime	Calcium hydroxide	Ca(OH)_2
52.	Milk of magnesia	Magnesium hydroxide	Mg(OH)_2
53.	Marsh gas	Methane	CH_4
54.	Malachite (green)	Basic copper carbonate	$\text{CuCO}_3 \cdot \text{Cu(OH)}_2$
55.	Magnesia	Magnesium oxide	MgO
56.	Magnesite	Magnesium carbonate	MgCO_3
57.	Mohr's salt	Ferrous ammonium sulphate hexahydrate	$(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$
58.	Muriatic acid	Hydrochloric acid	HCl
59.	Nessler's reagent	Potassium tetra-iodomercurate	$\text{K}_2[\text{HgI}_4]$
60.	Nitre	Potassium nitrate	KNO_3
61.	Nitre cake	Sodium hydrogen sulphate	NaHSO_4
62.	Nitrolim	Calcium cyanamide plus carbon	$\text{CaCN}_2 + \text{C}$
63.	Nitrate of lime (Nitrolime)	Basic calcium nitrate	$\text{Ca(NO}_3)_2 \cdot \text{CaO}$
64.	Oil of Vitriol	Concentrated Sulphuric acid	H_2SO_4
65.	Oleum	Pyrosulphuric acid	$\text{H}_2\text{S}_2\text{O}_7$
66.	Pearl ash	Potassium carbonate	K_2CO_3
67.	Philosopher's wool	Zinc oxide	ZnO
68.	Phosgene	Carbonyl chloride	COCl_2
69.	Potash	Potassium carbonate	K_2CO_3
70.	Potash alum	Potassium aluminium sulphate	$\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
71.	Plaster of Paris	Hemihydrate of calcium sulphate	$\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$
72.	Pyrogallol	Benzene-1, 2, 3-triol	$\text{C}_6\text{H}_3(\text{OH})_3$
73.	Pyrolusite	Manganese dioxide	MnO_2
74.	Quartz or silica	Silicon dioxide	SiO_2
75.	Quick lime	Calcium oxide	CaO
76.	Quick silver	Mercury	Hg
77.	Red lead	Triplumbic tetroxide	Pb_3O_4
78.	Rock salt	Sodium chloride	NaCl
79.	Rough	Ferric oxide	Fe_2O_3
80.	Sal ammoniac	Solid ammonium chloride	NH_4Cl
81.	Sal volatile	Solid ammonium carbonate	$(\text{NH}_4)_2\text{CO}_3$
82.	Salt cake	Anhydrous sodium sulphate	Na_2SO_4
83.	Salt petre	Potassium nitrate	KNO_3
84.	Siderite	Ferrous carbonate	FeCO_3
85.	Silica	Silicon dioxide	SiO_2
86.	Silica gel	Hydrated silica	$\text{SiO}_2 \cdot x\text{H}_2\text{O}$
87.	Slaked lime	Calcium hydroxide	Ca(OH)_2
88.	Soda ash	Anhydrous sodium carbonate	Na_2CO_3
89.	Water glass	Sodium dioxide containing excess of silica	$\text{Na}_2\text{SiO}_3 \cdot \text{SiO}_2$
90.	Super phosphate of lime	Mixture of calcium dihydrogen phosphate and calcium sulphate	$\text{Ca(H}_2\text{PO}_4)_2 \cdot \text{H}_2\text{O} + 2\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
91.	Verdigris	Basic copper acetate	$\text{Cu(CH}_3\text{COO)}_2$
92.	White lead	Basic lead carbonate	$2\text{PbCO}_3 \cdot \text{Pb(OH)}_2$
93.	White Vitriol	Heptahydrate of zinc sulphate	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
94.	Washing soda	Hydrated sodium carbonate	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
95.	Zincite	Zinc oxide	ZnO
96.	Zinc blende	Zinc sulphide	ZnS

VALENCY OF SOME COMMON BASIC AND ACIDIC RADICALS

Electrovalent positive ions (basic radicals)

Monovalent		Divalent		Trivalent	
1. Ammonium	NH_4^+	1. Argentite [Silver(II)]	Ag^{2+}	1. Aluminium	Al^{3+}
2. Aurous [Gold (I)]	Au^+	2. Barium	Ba^{2+}	2. Ferric [Iron (III)]	Fe^{3+}
3. Cuprous [Copper (I)]	Cu^+	3. Calcium	Ca^{2+}	3. Chromium	Cr^{3+}
4. Hydrogen	H^+	4. Cupric [Copper(II)]	Cu^{2+}	4. Auric [Gold (III)]	Au^{3+}
5. Mercurous [Mercury (I)]	Hg^+	5. Ferrous [Iron (II)]	Fe^{2+}	5. Bismuth	Bi^{3+}
6. Potassium	K^+	6. Magnesium	Mg^{2+}	6. Arsenic	As^{3+}
7. Sodium	Na^+	7. Mercuric [Mercury (II)]	Hg^{2+}	Tetravalent	
8. Argentous [Silver (I)]	Ag^+	8. Plumbous [Lead (II)]	Pb^{2+}		
9. Lithium	Li^+	9. Stannous [Tin (II)]	Sn^{2+}		
		10. Zinc	Zn^{2+}	1. Plumbic [Lead (IV)]	Pb^{4+}
		11. Platinous [Platinum (II)]	Pt^{2+}	2. Stannic [Tin (IV)]	Sn^{4+}
		12. Nickel	Ni^{2+}	3. Platinic [Platinum (IV)]	Pt^{4+}

ELECTROVALENT NEGATIVE IONS (ACIDIC RADICALS)

Monovalent		Divalent		Trivalent	
1. Acetate	CH_3COO^-	1. Carbonate	CO_3^{2-}	1. Arsenate	AsO_4^{3-}
2. Bicarbonate or Hydrogen carbonate	HCO_3^-	2. Dichromate	$\text{Cr}_2\text{O}_7^{2-}$	2. Nitride	N^{3-}
3. Bisulphide or Hydrogen sulphide	HS^-	3. Oxide	O^{2-}	3. Arsenite	AsO_3^{3-}
4. Bisulphate or Hydrogen sulphate	HSO_4^-	4. Peroxide	O_2^{2-}	4. Phosphide	P^{3-}
5. Bisulphite or Hydrogen sulphite	HSO_3^-	5. Sulphate	SO_4^{2-}	5. Phosphite	PO_3^{3-}
6. Bromide	Br^-	6. Sulphite	SO_3^{2-}	6. Phosphate	PO_4^{3-}
7. Chloride	Cl^-	7. Sulphide	S^{2-}	7. Borate	BO_3^{3-}
8. Permanganate	MnO_4^-	8. Silicate	SiO_3^{2-}	8. Aluminate	AlO_3^{3-}
9. Fluoride	F^-	9. Thiosulphate	$\text{S}_2\text{O}_3^{2-}$	Tetravalent	
10. Hydride	H^-	10. Zincate	ZnO_2^{2-}		
11. Hydroxide	OH^-	11. Plumbite	PbO_2^{2-}	1. Carbide	C^{4-}
12. Iodide	I^-	12. Stannate	SnO_3^{2-}	2. Ferrocyanide	$\text{Fe}(\text{CN})_6^{4-}$
13. Cyanide	CN^-	13. Manganate	MnO_4^{2-}		
14. Nitrate	NO_3^-	14. Chromate	CrO_4^{2-}		
15. Nitrite	NO_2^-	15. Oxalate	$(\text{COO})_2^{2-}$		
16. Chlorite	ClO_2^-				
17. Hypochlorite	ClO^-				
18. Chlorate	ClO_3^-				
19. Perchlorate	ClO_4^-				
20. Meta aluminate	AlO_2^-				